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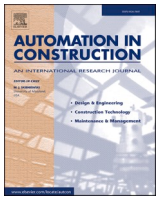

Efficient low-collision UAV-based automated structural surface inspection using geometric digital twin and voxelized obstacle information

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ABSTRACT

The application of Unmanned Aerial Vehicle (UAV) automatic flight is increasingly popular for structural surface inspection. To address the low level of automation and insufficient adaption of the flight path in response to environmental obstacles, a method of automatic planning UAV inspection mission based on the Geometric Digital Twin (GDT) model and Voxelized Obstacle Information (VOI) is proposed. First, a method for shifting the Field of View (FOV) centroids in parallel is proposed to efficiently generate inspection waypoints. Second, a waypoints adjustment method based on environmental VOI of 3D point clouds is proposed to address the safety issues. Third, a method combining Genetic Algorithm (GA) with A* based on VOI is proposed for optimizing UAV flight path to avoid real-world obstacles. The feasibility of the proposed methods was verified in both an office building and a steel truss bridge. Compared to existing methods, the efficiency is significantly improved.

1. Introduction

Periodic structural surfaces inspection is crucial for ensuring the structural safety of large-scale infrastructures such as buildings and bridges. Traditionally, such inspection has been largely conducted manually, which is time consuming and prone to errors. More recently, due to the agility and efficiency of Unmanned Aerial Vehicles (UAVs) in capturing surface images, some studies have begun to apply UAV technology for structural surface inspection and shown promising results [1–3]. This process typically involves orienting the UAV camera towards the target structure where it captures inspection images of the visible coverage area on the structural surface, which is defined as the Field of View (FOV). During the process of using UAVs to fly around structures and acquire inspection image data, three main control indicators are emphasized [4–6]: (1) safety, to ensure that UAVs do not collide with obstacles; (2) comprehensiveness, to acquire complete images of the whole surface to avoid missing inspection; (3) efficiency, to avoid unnecessary flight burdens. When using traditional manual UAV flight, such multiple and possibly conflicting goals can be challenging to be achieved simultaneously, especially for large-scale and complex structures [7–9]. To tackle this issue, some researchers have comprehensively reviewed and studied the pre-planning of rational UAV flight missions to enable automatic execution [10–12]. Compared to the manual control method, it has been shown that pre-planning can significantly enhance the safety and efficiency of the inspection [6]. For an effective pre-planning of flight mission, the first step is to generate suitable structural inspection waypoints, which are the specified location coordinates where a UAV hovers and acquires images.

Properly choosing of waypoints is crucial for the complete while economical covering of the structural surface. Naturally, a reference model to the real structure is necessary for determining the distribution of the waypoints. Such a reference model can be classified into five levels of detail (LoD) according to the Open Geospatial Consortium (OGC) standard for City Geography Markup Language (GML) 2.0 [13]. A building example is used to illustrate the different levels from LoD0 to LoD4, as shown in Fig. 1. The LoD0 model [14–16] represents the structural projection on a 2D plane, which is typically acquired using footprint extraction methods [17,18]. Such a simple model can be used for the generation of waypoints that focus on the acquisition of rough top view images. To generate waypoints that focus on the acquisition of both rough facade and top view images, the LoD1 model [19], which is a block model derived by extending the LoD0 model vertically, can be used. Although in 3D, the LoD1 model lacks differentiated facades and the roof structures [20], which can create difficulties for structures with

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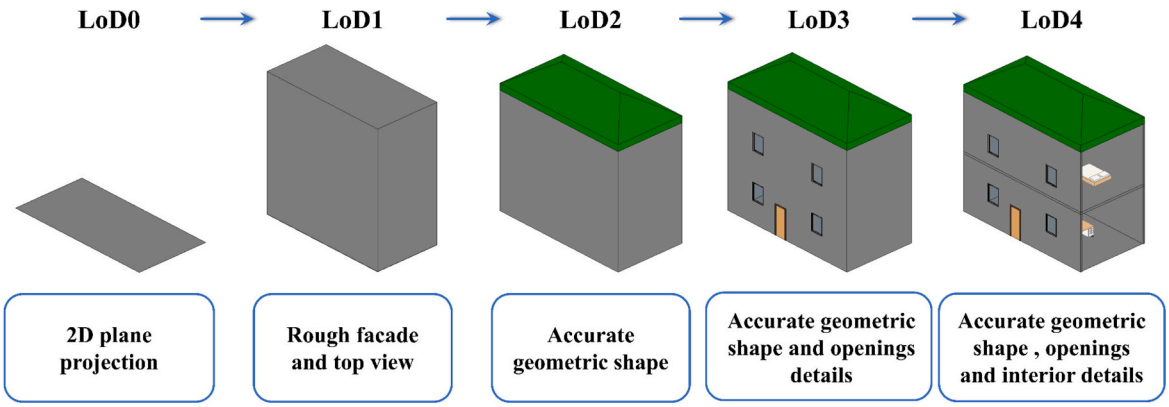


Fig. 1. Five building levels of detail from LoD0 to LoD4 (created based on [13]).

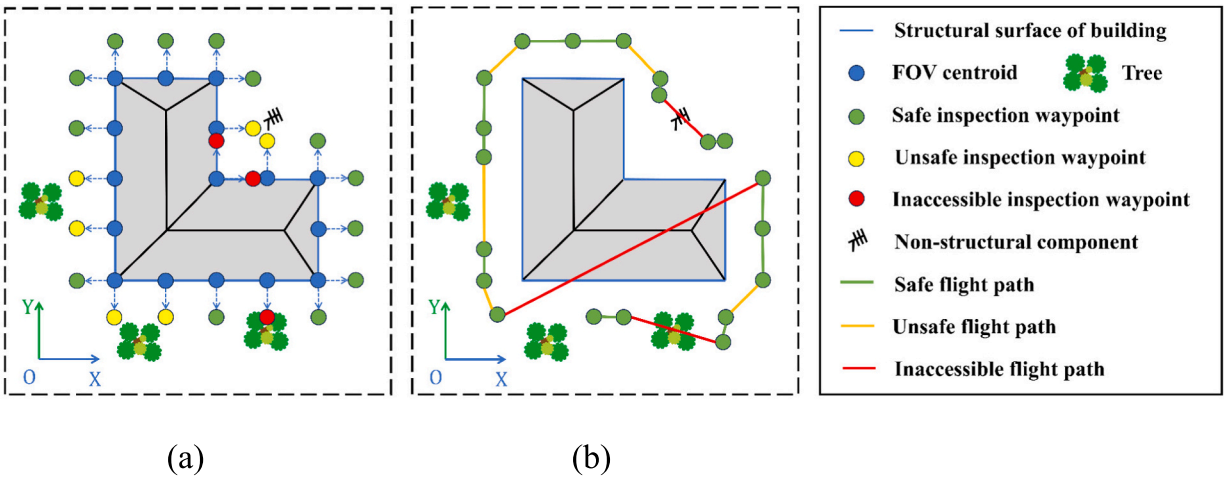


Fig. 2. Issues of waypoints generation due to parallel shift strategy: (a) Unsafe or inaccessible inspection waypoints, (b) Unsafe or inaccessible flight path.

irregular shapes. To deal with this, the LoD2 model, in which the accurate geometric shape of the structure is represented, can be used. With openings in the structure, such as doors and windows, the LoD2 model can be further upgraded to LoD3 [21]. By further including details of the structural interior, the LoD4 model can be created [6]. From their characteristics it can be seen that it is challenging to use LoD0 and LoD1 models for effective structural surface inspection, while LoD2, LoD3 and LoD4 can serve as better reference models. Previously, some studies have generated waypoints based on models constructed by design drawings defined under LoD3, such as manually constructed Building Information Modeling (BIM) [22–25] and 3D geometric model [26,27]. While these models can serve as effective reference models can be automatically generated through methods such as deep learning [28] and progressive model-driven approach [29], their openings and internal details are not essential information for external surface inspection waypoint generation. Moreover, the large amount of training data and modeling complexity required do hinder the automatic construction of the efficient reference models. Thus, it can be concluded that the LoD2 model is the most suitable choice for the task of external surface inspection.

Some studies have utilized 3D point clouds to extract the geometric shape of the structure [4,30–32], voxelized it or adapt observation planes to the geometry of the scene for waypoints generation and iteratively optimizing the number and positions of waypoints based on data quality [33]. The 3D point clouds can be produced by depth devices [34,35] and multi-view reconstruction [36], and extensive research has been conducted on the methods of point clouds generation [37–39] and factors affecting the quality of point clouds [11,12,40]. Processing 3D point clouds and voxelized models can effectively generate inspection waypoints. However, due to the inclusion of significant non-structural components and surrounding environments, it becomes challenging to focus on the structural surface. Additionally, since these methods involve a large amount of discrete data, they tend to be less accurate and efficient compared to waypoints generation using LoD2 models.

Several studies have addressed waypoints generation based on reference models. Tan et al. [22] proposed a “three-steps” region division method. First, the polygonal surface of the building is divided into several rectangular regions, with the corner point of each rectangular region serving as initial point to create the first rectangle primitive. These primitives are then translated until them cover all rectangular regions on the surface, and the offset of each primitive’s center point is extracted to generate inspection waypoints for the surface. Additionally, by pre-dividing the plane into grids, a polygonal surface can be directly divided in a “two-steps” process. Yang et al. [41] used UAVs to collect overhead inspection images of debris fans, manually inputting the boundary information of the debris fan into the pre-divided grid and generating waypoints based on the grid’s center points, effectively covering the irregularly shaped debris fan. Yu et al. [15] also conducted research on the generation of overhead waypoints, using a four-neighborhood growing method to determine the areas to be covered and dividing the grid to generate waypoints. However, further research is needed on these methods to adapt them to the “two-steps” division of 3D polygonal surfaces of GDT model, thereby improving the efficiency of waypoints generation.

A subsequent task after obtaining the waypoints is to adjust them to address potential safety issues. Such issues, which are often overlooked

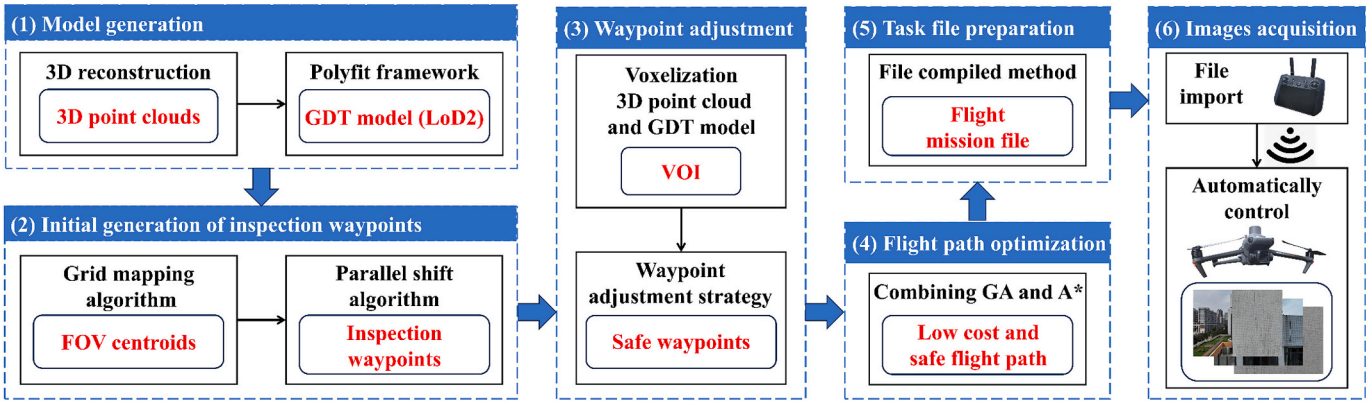


Fig. 3. Technology roadmap of automatic UAV inspection.

in previous research, arise from the nature of the commonly used waypoints generation methods and the unavoidable waypoints conflicting with real-world environmental obstacles, i.e., parallel shift strategies and obstacles. In these methods, feature points are first extracted from 3D point clouds' plane and boundary [4,31,32], the center points of triangle mesh [30], or the sub region centroids of BIM surface [22,23,25], which are then shifted outward in parallel to determine the waypoints. As illustrated in Fig. 2(a), waypoints generated with such strategies may be within obstacles, such as trees, power lines surrounding the building, and non-structural components installed on the building, or too close to them. Furthermore, the flight paths planned in these manners reduce flexibility and are less than optimal [42]. Therefore, it is necessary to adjust the positions of the waypoints according to the obstacle information around the structure to prevent unsafe or inaccessible waypoints. Some studies have considered the environmental obstacles in the process of flight path planning. For instance, Sun et al. [43] used a semantic segmentation method to divide obstacle areas in orthophotos, providing references for UAV flight path planning at the 2D level. Some other studies have considered manually constructed obstacle models at the 3D level for path planning [44,45]. Tan et al. [46] researched multi-UAV path planning issues using virtual obstacles that simulate the environment. Obstacle information can also be expressed through automatic methods. Phung et al. [4] expressed the obstacle information using the maximum bounding boxes of 3D point clouds, ensuring that waypoints and flight paths avoid obstacles. While these methods can lower the flight risks to a certain degree by considering the environment obstacles, manually constructed virtual obstacle models may differ from actual conditions and be inefficient. Additionally, using rough models such as bounding boxes to represent environmental obstacles may result in the loss of valuable accessible areas, especially for inspecting complex structures. Therefore, it is vital to further consider more accurate obstacle information of the real-life environment when using the parallel shift strategy.

Once the ultimate waypoints are determined, a flight path optimization shall be conducted to find the optimal route that sequentially connects all generated unordered inspection waypoints. A typical unsafe or inaccessible flight path is shown in Fig. 2(b), in which parts of it pass through obstacles or being too close to obstacles. It is necessary to optimize the flight path to ensure UAV flight safety while also minimalize flight burden. These optimization challenges bear resemblance to the Traveling Salesman Problem (TSP) and are commonly addressed using heuristic algorithms, such as Genetic Algorithm (GA) [47], self-adaptive genetic simulated annealing algorithm (AGASA) [2], and Ant Colony Optimization (ACO) [15]. Research on these algorithms focuses on improving their execution speed, convergence, and avoiding local optima. These methods can enhance the accuracy and efficiency of path optimization. However, the issues of obstacle avoidance and energy consumption in UAV during practical engineering applications also warrant further attention. Optimization of flight path that also tackles safety issues is a constrained optimization problem, in which the considerations of obstacles as constraints in heuristic algorithms are incorporated. Koch et al. [48] used semantic maps overlaid on 3D model to determine if there are restrictions on UAV flight along these paths, and provided the results to heuristic algorithms to optimize the flight path. In environments where obstacles cannot be completely avoided using heuristic algorithms alone. To address this issue, some studies use a combination of automatic path finding algorithms and heuristic algorithms to ensure that the optimized flight paths avoid obstacles [4,5,23]. Among these, Phung et al. [4] considered UAV energy consumption related to flight direction when planning paths using such methods. Furthermore, Tan et al. [23] not only planned UAV flight paths but also devised routes for the UAV to reach designated locations for battery replacement. However, it is vital to ensure that flight paths maintain a certain safe distance from real-world environmental obstacles and reduce costs associated with UAV turns due to obstacle avoidance, which requires further research. In other words, it is crucial to consider more real and accurate obstacle information and minimize transitional waypoints for turns as much as possible to optimize the flight path, which has not been sufficiently considered in the past studies.

To tackle these issues, this paper proposes a novel method for automatically planning UAV inspection flight missions for capturing surface images based on Geometric Digital Twin (GDT) model and Voxelized Obstacle Information (VOI). First, a LoD2-class GDT model, which can be automatically and rapidly constructed [49–51], is innovatively applied to the inspection waypoints generation. Based on the accurate geometric shape of the GDT model, a preliminary waypoints generation method that involves shifting FOV centroids in parallel is proposed, enhancing the automation and efficiency of the waypoint generation process. Second, a method utilizing more precise VOI for inspection waypoints adjustment is proposed. This method employs voxelization and regional division of the space containing the 3D point clouds and GDT model to relocate unsafe or inaccessible waypoints to safer areas. Finally, a heuristic path optimization method combining GA with A* based on VOI is proposed to define the safe flight path, incorporating consideration of real-world environmental obstacles and reducing the additional energy and time costs associated with automatic routing through turn penalties. The method proposed in this paper efficiently and effectively generates waypoints based on the GDT model and ensures the safety of both the waypoints and the flight path connecting them through VOI. This comprehensive 'point-to-line' approach safeguards the safety of UAV in all aspects, ultimately acquiring effective 'surface' imagery data. The main content of this paper is structured as follows: the proposed method is presented in Section 2. In Section 3, the method is applied to a building and a bridge to demonstrate the feasibility of the proposed method. The final section presents the

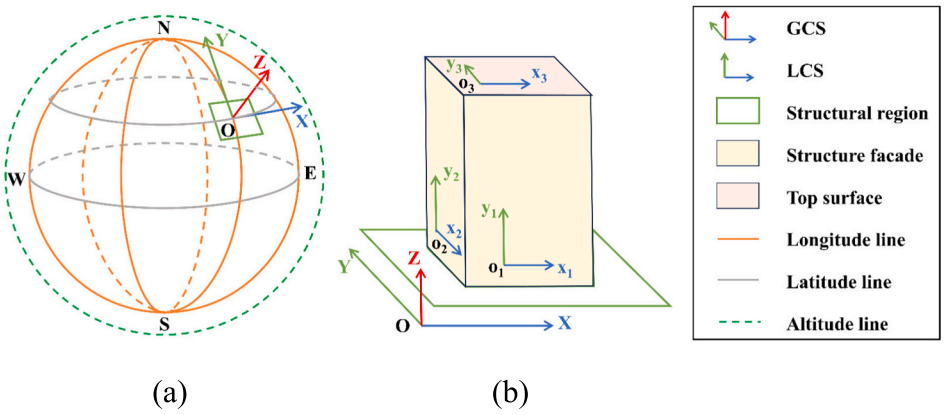


Fig. 4. Coordinate systems used: (a) GCS and WGS-84, (b) GCS and LCS.

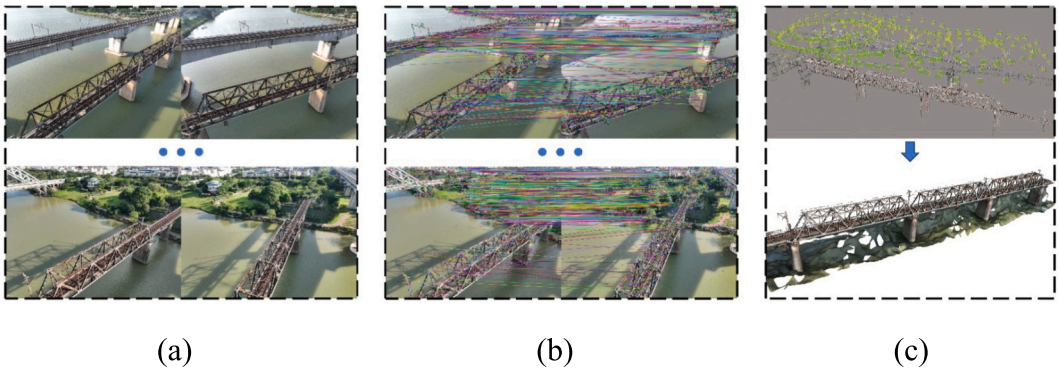


Fig. 5. Process of multi-view 3D reconstruction: (a)multi-view images acquisition, (b) Detection and matching of feature points in multi-view images, (c) Sparse and dense point clouds reconstruction.

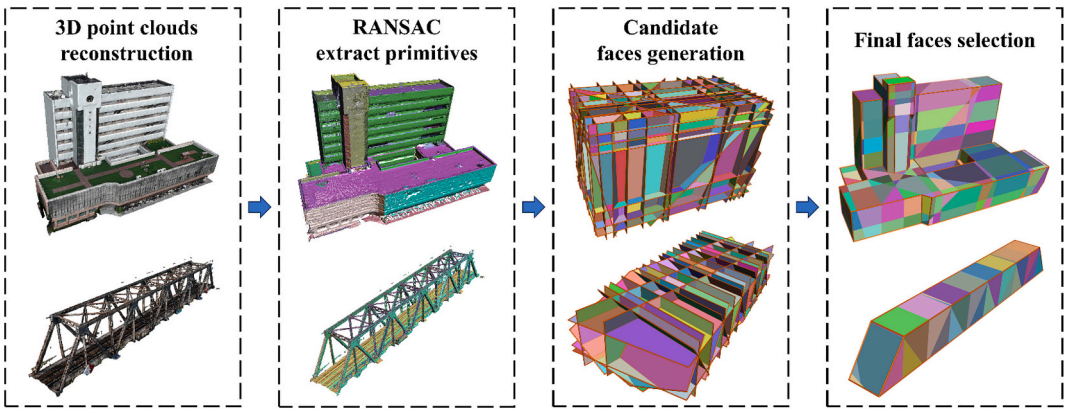


Fig. 6. Process of structural GDT model generation based on Polyfit framework [49].

conclusions.

2. Proposed UAV automatic inspection method using GDT model and VOI

This paper proposes an automatic planning method for UAV inspection missions based on the GDT model and VOI, allowing for a rapid and automatic deployment of UAV inspection flight mission, which can achieve automatic and safe inspection of structure surfaces. The main procedure of the method includes: (1) automatically constructing a GDT model based on 3D point clouds; (2) automatically generating preliminary inspection waypoints based on the GDT model; (3) adjusting the preliminary inspection waypoints based on divided areas that reference VOI; (4) optimizing the UAV flight path using the method combining GA with A* based on VOI; (5) compiling UAV flight files to realize automatic inspection. The technology roadmap is shown in Fig. 3.

In this paper, as shown in Fig. 4, three coordinate systems are used in the presentation of the proposed method. The first is the 3D Global Coordinate System (GCS), OXYZ, where the positive direction of the X-axis points to true east, the positive direction of the Y-axis points to true north, and the positive direction of the Z-axis is determined by the right-hand rule, with O as the origin. The second is the 2D Local Coordinate System (LCS), oxy, located on the surface of the GDT model. On the

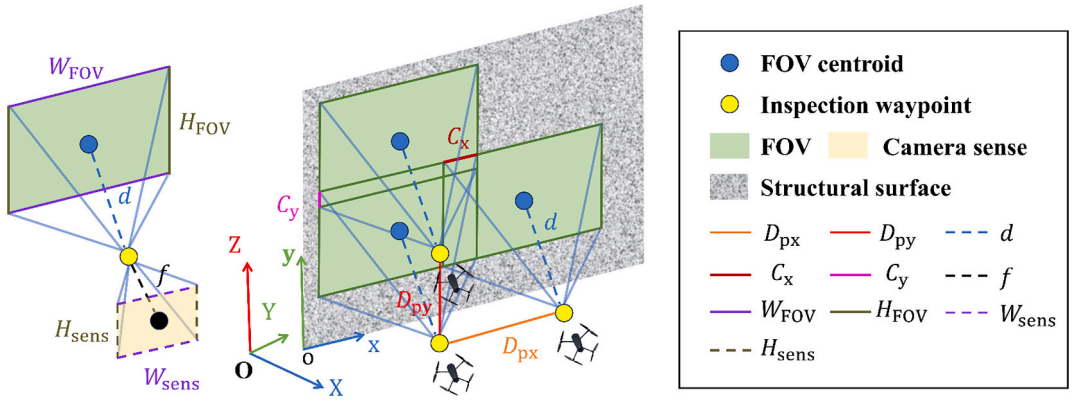


Fig. 7. Definition for key parameters of waypoints.

facade of the structure, the positive direction of the y-axis in the LCS points upward, and on the top surface of the structure, the positive direction of the y-axis is perpendicular to the longest edge of that surface, with the positive direction of the x-axis determined by the right-hand rule. The third is the World Geodetic System-1984 (WGS-84), "Latitude Longitude Altitude" (LLA), which is used for the compilation of the flight mission files.

2.1. Automatic GDT model construction based on 3D point clouds

As aforementioned, a reference model is the basis for generating the waypoints and an automatically constructed GDT model based on 3D point clouds is used as the reference model for the proposed method. In this study, 3D point clouds are produced through a multi-view 3D reconstruction (as shown in Fig. 5), which is of high efficiency and low cost [37], and can represent the actual position information of objects in the real world. Although there may be some potential positioning errors, these can be effectively limited by increasing the number of Ground Control Points (GCPs) [39]. The process consists of multi-view images acquisition and reconstruction stages. The image acquisition can be done manually [36] or automatically at a low cost [11]. The reconstruction process involves Structure from Motion (SFM) and Multi-View

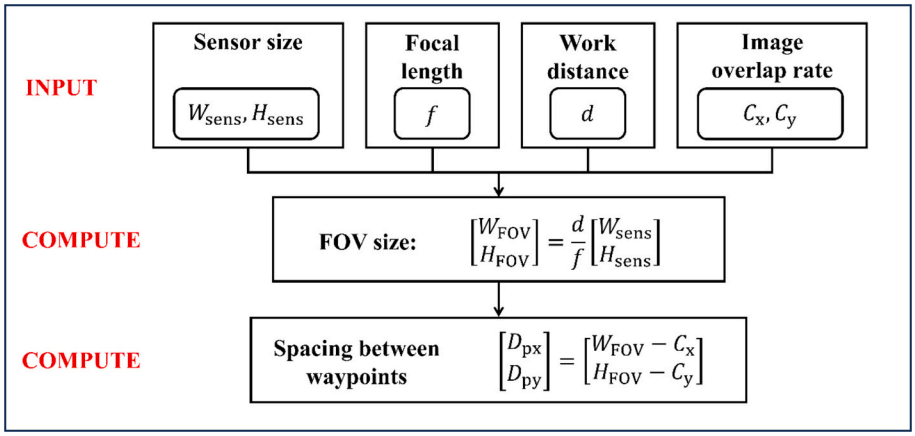


Fig. 8. Process of determining waypoints parameters.

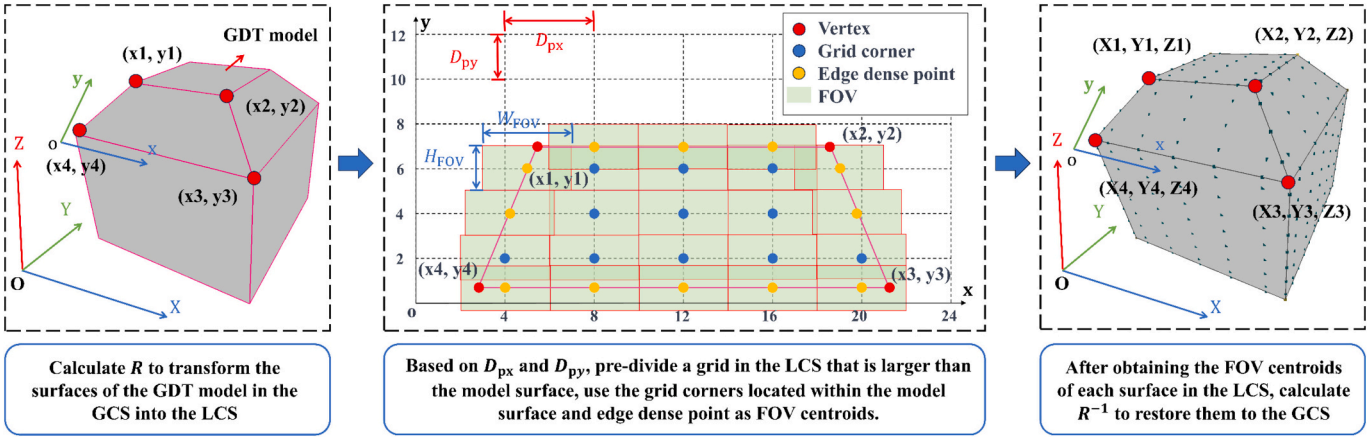


Fig. 9. Process of FOV centroids generation based on grid mapping method.

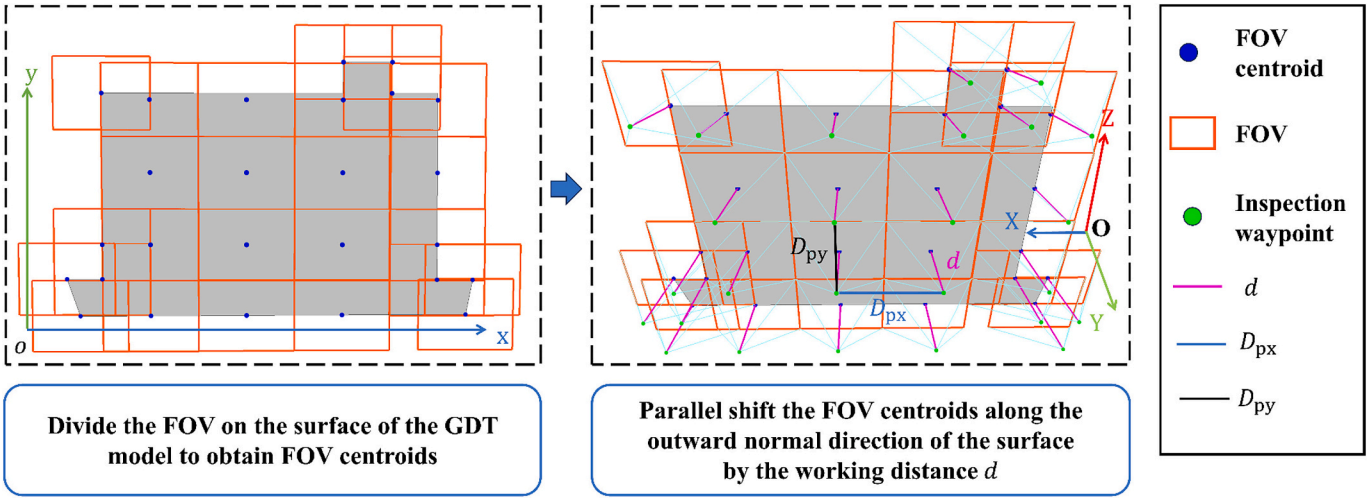


Fig. 10. Process of inspection waypoints generation based on parallel shift method.

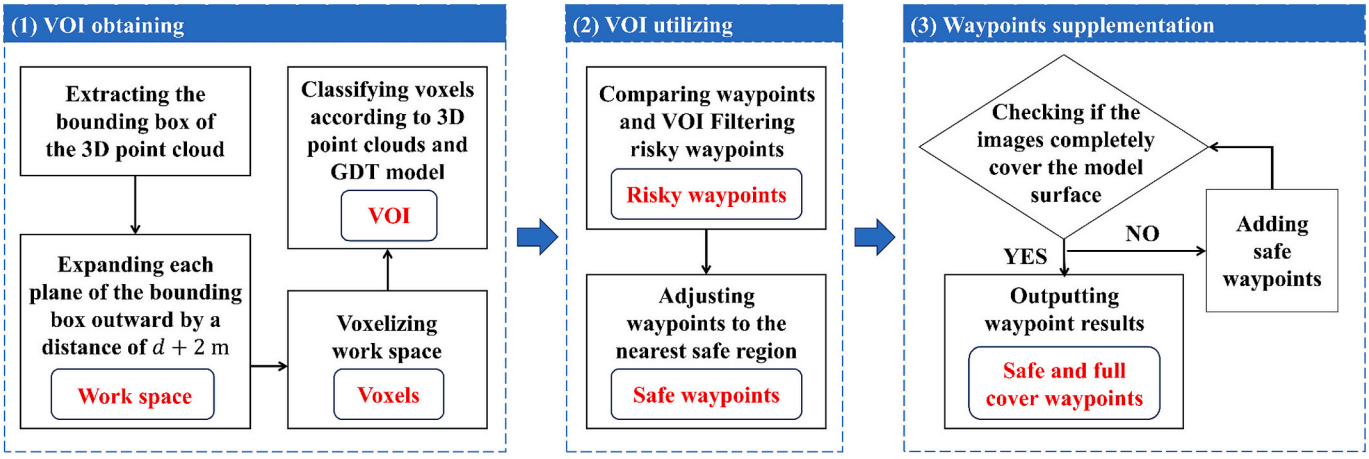


Fig. 11. Procedure of waypoints adjustment.

Stereo (MVS) components [9,52,53]. In this process, SFM extracts and matches feature points from the acquired multi-view images, generating a sparse point cloud, while MVS estimates the depth maps based on the sparse point cloud and further generates a dense point cloud in 3D space.

With the 3D point clouds, the GDT model can be automatically constructed based on the Polyfit framework proposed by Nan et al. [49]. In Fig. 6, the procedure of the construction of the GDT model is illustrated with a building and a bridge: (1) extracting the plane primitives based on Random Sample Consensus (RANSAC) [54], and sampling points in the point cloud that belong to the same plane; (2) generating candidate faces for the GDT model based on the sampling results; (3) selecting the final faces of the GDT model based on topological rules.

After the generation of the model, it is necessary to assess its feasibility and accuracy. The Root Mean Square Error (RMSE) is adopted for this purpose, where the 3D point clouds are considered as actual observations, and the GDT model as predictions. The fitting error between the GDT model and the 3D point clouds is calculated using Eq. (1) [50].

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{i=1}^{T}(\Delta D_i)^2} \tag{1}$$

where, T is the total number of points in the 3D point clouds; ΔD_i is the Euclidean distance between each point and the nearest surface of the GDT model.

To provide a more direct assessment of the fit between the computed RMSE and the structural 3D point clouds, the RMSE is compared against the structural volume. Define L as the side length of a cube with a volume equal to that of the structure, representing the structural volume. When RMSE $<0.02\ L$, the fit between the GDT model and the 3D point clouds is considered good; when $0.02\ L <$ RMSE $<0.05\ L$, the fit is considered moderate; and when RMSE $>0.05\ L$, the fit is considered poor. Numerical evaluation is only one criterion for assessing the fit; a more comprehensive judgment can be made by directly overlaying the GDT model with the 3D point clouds for visual inspection.

2.2. Inspection waypoints generation based on FOV centroids' parallel shift method

After the GDT model is generated, the FOVs on its are pre-divided using the proposed grid mapping algorithm based on the camera's FOV. Subsequently, using the proposed parallel shift algorithm, the FOV centroids on the GDT model surface are parallelly shifted along the surface normal up to the working distance d of the camera, achieving the generation of preliminary inspection waypoints, which also serve as the camera's center points. The images captured by the UAV when it reaches these waypoints correspond to the subregions previously delineated on the GDT model surface.

2.2.1. Waypoint parameters calculation method

Each waypoint has a corresponding FOV on the GDT model surface.

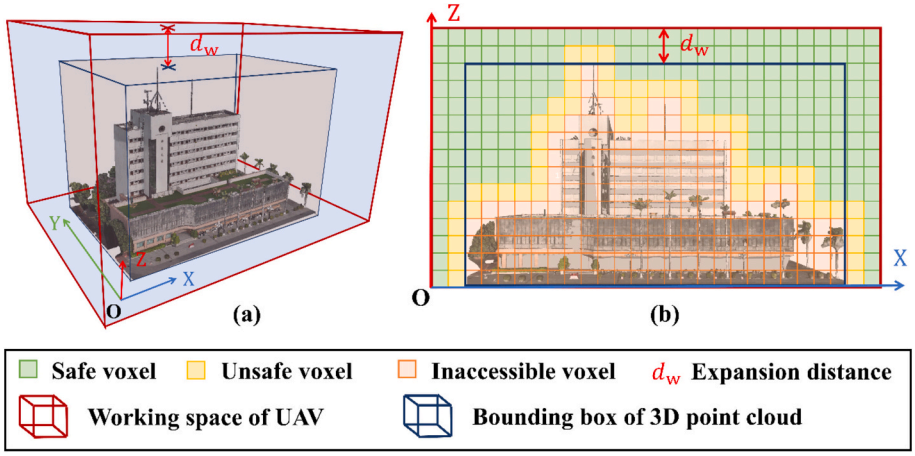


Fig. 12. Process of UAV inspection working space voxelization and areas division: (a) Extending bounding box to define workspace, (b) Voxelizing workspace and classifying voxels and spatial region to get VOI.

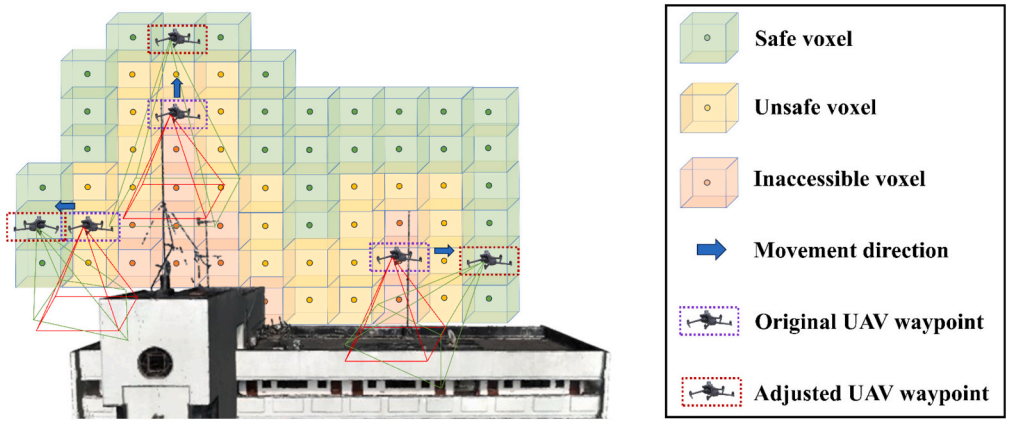


Fig. 13. Moving adjustment of waypoints in inaccessible or unsafe areas.

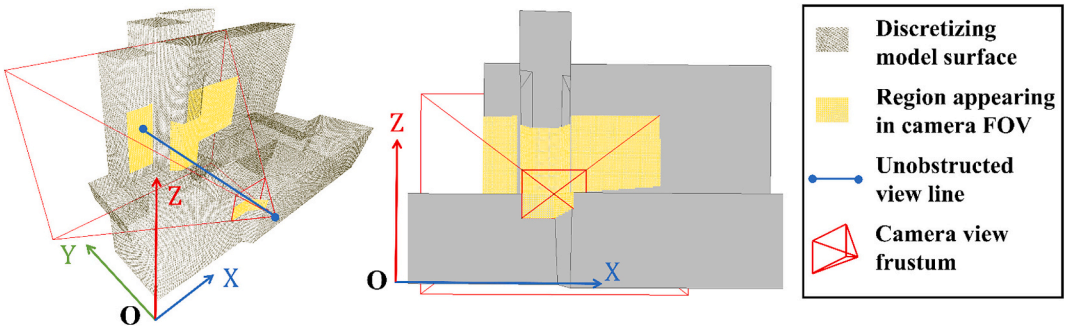


Fig. 14. Calculation of structural surface area each camera covers.

To determine the FOV so as to achieve complete covering of the model surface, several waypoint parameters need to be defined and calculated, the relationships between waypoint parameters are as shown in Fig. 7. They include the camera sensor width W_{sens}, camera sensor height H_{sens}, camera focal length f, working distance d, FOV width W_{FOV}, FOV height H_{FOV}, FOV horizontal overlap width C_x, FOV vertical overlap height C_y, horizontal spacing between waypoints D_{px}, and vertical spacing between waypoints D_{py}. Among these, W_{sens}, H_{sens}, f are known camera parameters, and d can be determined by planners based on the actual conditions of the UAV autonomous inspection and targets on the structure surface. The larger of d,the larger the area of FOV, and the fewer waypoints needed to cover the structure surface [23]; however, the images captured at each waypoint are more likely to lose details. Therefore, for detecting small surface damages, d should be appropriately reduced to ensure sufficient resolution. Additionally, the value of d should also be related to the flight safety of the UAV. The working scenario of the UAV should also be considered; for instance, in dense urban areas, d should not be too large to prevent collisions with surrounding buildings, whereas in open areas such as inspecting bridges over rivers, d can be appropriately increased. However, to meet the detailed requirements of the inspection images, d should not exceed 10 m. In this paper, it is suggested that d should be at least 5 m to ensure

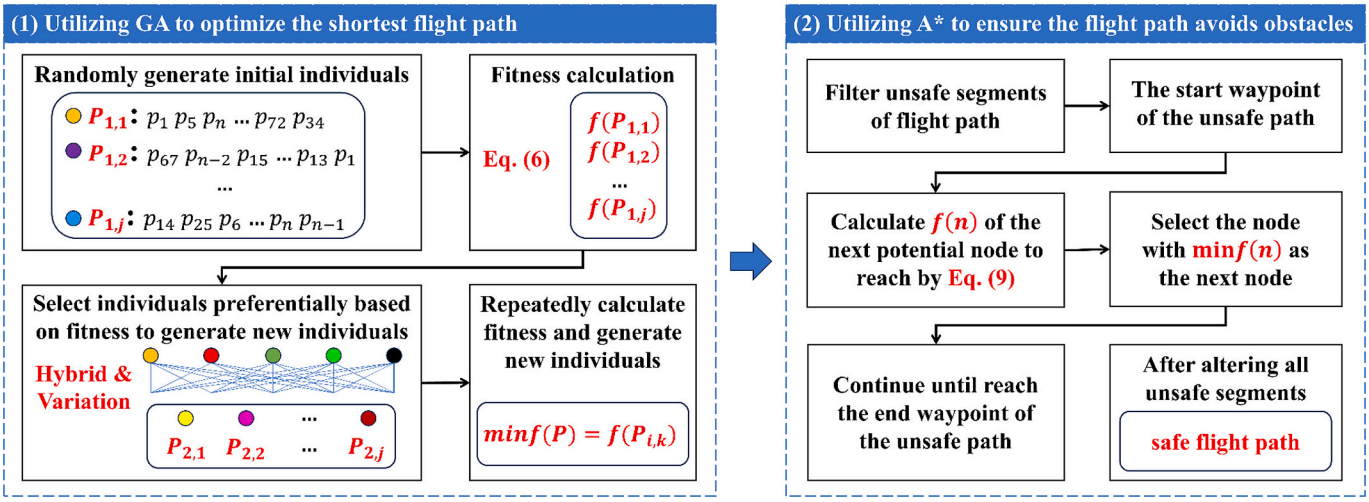


Fig. 15. Procedure of flight path optimization.

that UAV maintains an appropriate initial safety distance from the structure surface. Based on the known camera parameters and d, W_{FOV} and H_{FOV} can be further calculated by Eq. (2) [22].

$$\begin{bmatrix} W_{FOV} \\ H_{FOV} \end{bmatrix} = \frac{d}{f} \begin{bmatrix} W_{sens} \\ H_{sens} \end{bmatrix} \tag{2}$$

Among the parameters of the waypoints, the values C_x and C_y depend on the user requirements for capturing target images. For damage inspection, the values of C_x and C_y should be 0 [4] to meet the requirements for image redundancy reduction, they can also be set to appropriate values to compensate the impact of UAV positioning errors on the actual FOV of the camera. Based on C_x, C_y, W_{FOV}, H_{FOV}, D_{px} and D_{py} can be calculated by Eq. (3).

$$\begin{bmatrix} D_{px} \\ D_{py} \end{bmatrix} = \begin{bmatrix} W_{FOV} - C_x \\ H_{FOV} - C_y \end{bmatrix} \tag{3}$$

As shown in Fig. 8, after the aforementioned determination and calculations, the key waypoints parameters, including d, D_{px} and D_{py}, are obtained. These key parameters will serve as inputs to the subsequent waypoints generation algorithm.

2.2.2. Grid mapping method of dividing FOVs for obtaining the centroids

After the parameters of the waypoints are determined and calculated, the FOVs can be divided on the surfaces of the model, allowing the waypoints generation based on the FOVs' centroid. Previous studies have proposed considering the model surface as a set and the FOVs as subsets within the set, completing the division of the model surface's FOV by determining the subsets [4]. Covering regular rectangular planes with the FOVs is relatively straightforward; however, many buildings and bridges have surfaces that are irregular polygonal planes. Irregular polygonal planes can also be used to fit curved surfaces and detail features within structure. For such planes in 3D space, Tan et al. [22] employed the "three-step" method described in Section 1, while Yu et al. [15] used the "two-step" method for dividing FOVs on polygonal planes in 2D space. Inspired by these methods, a grid mapping algorithm (see Fig. 9) is proposed to improve efficiency by achieve a "two-steps" division of the FOVs on the polygon surface of the GDT model in the 3D space. In this algorithm, an area larger than the model surface is pre-divided into FOVs in the local coordinate system. The model surface is then transformed from the global coordinate system to the local coordinate system. Based on the polygonal surface of the model, FOVs that fully cover the surface are automatically selected. This approach allows for the segmentation of FOVs on irregular polygonal planes and facilitates automatic and efficient operations.

As shown in Fig. 9, the 3D spatial coordinate system in which the GDT model resides is GCS, and the 2D planar coordinate system parallel to the surfaces of the GDT model is LCS. The rotation matrix R of each surface for transforming from GCS to LCS can be calculated using Eq. (4) [55]. Multiply the coordinates of the vertices of each model surface in the GCS by R, to transform them into the LCS.

$$R = \begin{bmatrix} \beta x^2 + cos\theta & \beta xy - zsin\theta & \beta xz + ysin\theta \\ \beta xy + zsin\theta & \beta y^2 + cos\theta & \beta yz - xsin\theta \\ \beta xz - ysin\theta & \beta yz + xsin\theta & \beta z^2 + cos\theta \end{bmatrix}, \ \beta = 1 - cos\theta \tag{4}$$

where, $cos\theta$ and $sin\theta$ are the cosine and sine of the angle between the outer normal of the polygonal plane of the GDT model in GCS and the Z-axis, respectively; x, y and z are the values of the plane's rotation axis vectors, respectively.

Then, in the LCS, a grid with a width of D_{px} and a height of D_{py} is divided. The range of the grid is set according to the volume of the target structure and should be larger than the area of each surface of the GDT model. As described in Subsection 2.2.1 regarding the relationship between waypoint spacing and FOV, the grid corners within the model surface can serve as FOV centroids. To ensure complete coverage of the model surface, the vertices of the model surface and additional densified points along the edges are also used as FOV centroids. The principle for densifying edge points is that if the intersection of the grid line and the edge is within $D_{px}/2$ in the x- or y-direction from a grid corner, the intersection point should not be used as an FOV centroid; otherwise, it should be used as one. Additionally, redundancy can occur at the corners of the model surface if the same model surface vertex is used as an FOV centroid along with nearby grid corners or densified edge points also serving as FOV centroids. To address this, a redundancy distance d_b is set. If the distance between two FOV centroids is less than d_b, one of the FOV centroids is removed. The redundancy distance d_b can be set between $\frac{1}{10} \sim \frac{1}{5} min(D_{px}, D_{py})$. This method of appropriate densification ensures complete coverage without dividing redundant FOVs. Finally, calculate the inverse matrix R^{-1} of the rotation matrix R, and multiply the FOV centroids coordinates by R^{-1} to restore the FOV centroids located in the LCS back to the GCS. This process achieves the division of FOV on each surface of the GDT model and obtains the FOV centroids.

2.2.3. Generating inspection waypoints based on FOV centroid's parallel shift

The grid mapping method can determine the FOV centroids on the surfaces of the GDT model. Based on the proposed parallel shift algorithm (see Fig. 10), these centroids are shifted along the surfaces' outer normal vectors by d, thereby generating UAV inspection waypoints. The

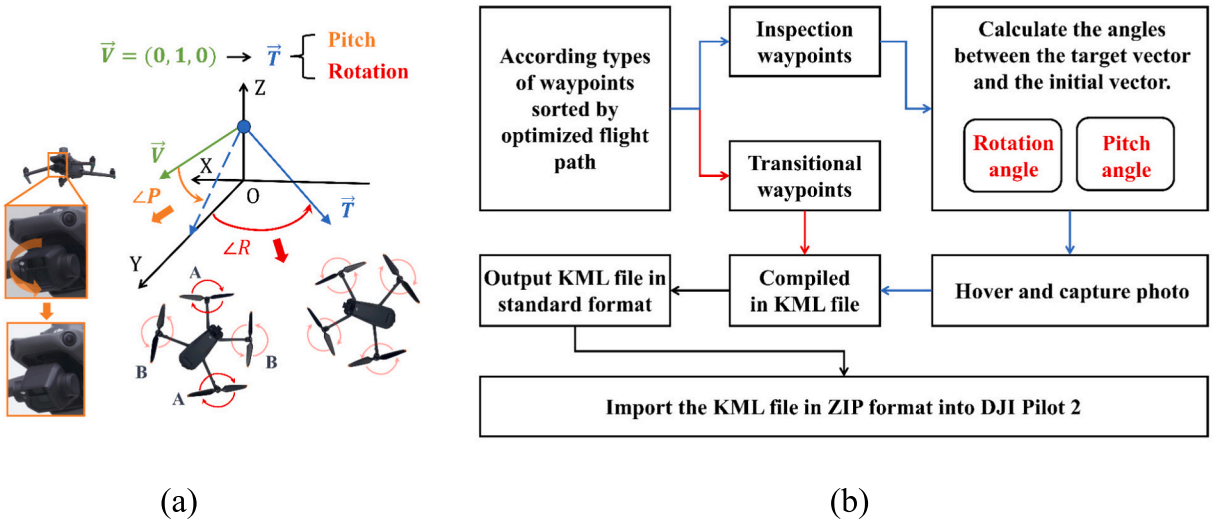


Fig. 16. File compilation method for UAV inspection mission: (a) UAV's camera orientation adjustment method, (b) Process of mission file compilation.

normal vectors for each surface of the GDT model can be calculated using Eq. (5).

$$\overrightarrow{n} = (V_{i+2} - V_{i+1}) \times (V_{i+1} - V_i) \tag{5}$$

where, $\overrightarrow{n}$ is the normal vector of the polygonal plane of the GDT model; V_{i+2}, V_{i+1} and V_i are any three non-collinear vertices on the polygonal plane.

2.3. Inspection waypoints adjustment method based on VOI

As mentioned in Section 1, because the bulk-generated waypoints do not account for the impact of environmental obstacles, they may be in unsafe or inaccessible positions, potentially lead to collisions. As summarized in Section 1, previous studies that generated waypoints based on design models often overlooked the impact of real environmental obstacles on UAV flight safety [5,22–24]; other studies represented environmental obstacles with rough bounding boxes [4], which can still miss some valuable accessible areas. On the other hand, 3D point clouds used in this study, as reconstructions of real scenes, can accurately depict the positions of environmental obstacles. Voxels, as subdivisions of 3D space [56], can clearly express the geometric information of environmental obstacles. Therefore, after voxelizing the 3D point clouds used to establish the GDT model as mentioned in Subsection 2.1, the VOI is obtained. A method for adjusting inspection waypoints utilizing more precise VOI to refine waypoints adjustment is then proposed.

As shown in Fig. 11, the process is as follows: firstly, based on the maximum and minimum values of points in the 3D point clouds along each coordinate axis, a cubic bounding box for the 3D point clouds is constructed, and all surfaces of the bounding box except the bottom are expanded outward by d_w to create the working space of UAV, which contains the waypoints. As shown in Fig. 12, the larger the expansion distance d_w, the larger the working space of the UAV. With the same voxel size, more voxels are needed to fill the working space, and the computational cost for obtaining the VOI increases. Therefore, this paper considers an expansion distance d_w slightly greater than the working distance d, which as mentioned in Subsection 2.2.1, specifically $d_w = d + 2$ m, to balance computational efficiency with sufficient adjustment space for the waypoints. Subsequently, the voxel size S is set, and the working space is voxelized. The choice of S affects both computational efficiency and VOI precision. For the inspection of structures with smaller volumes, it is advisable to choose a smaller S, but it should be kept within the range of 0.5 m to 1 m to ensure the precision of the VOI. Based on the 3D point clouds and the GDT model, all voxels within the working space are classified into three categories (details of the voxel classification are as shown in Fig. 13): the first category is the inaccessible voxels, which are considered unreachable for UAV if the voxel contains points from the 3D point clouds or if the center point of the voxel is inside the GDT model; the second category is the unsafe voxels, which is considered having potential dangerous posed for UAV

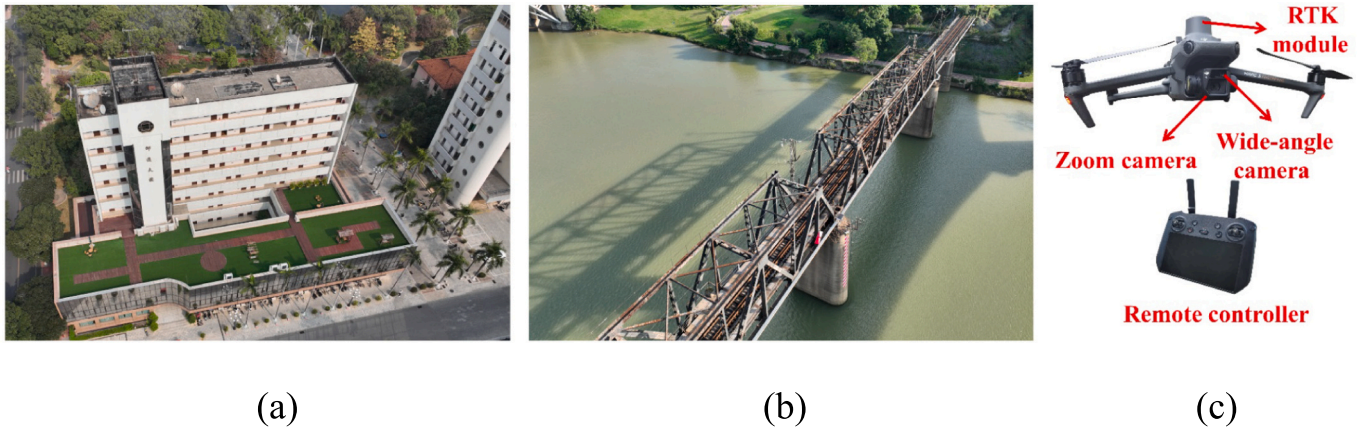
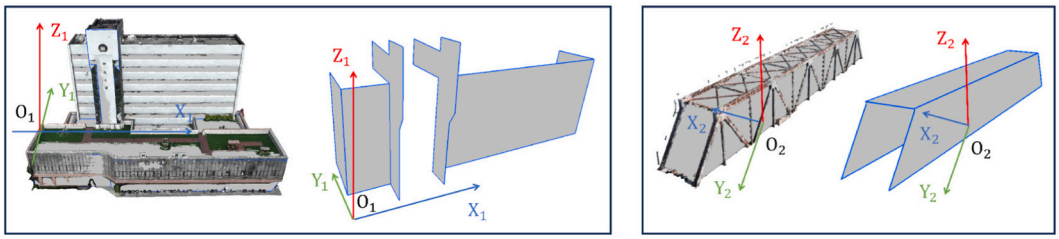


Fig. 17. Field test structures and equipment: (a) Office building, (b) Steel truss bridge, (c) UAV and remote controller.



(a) (b)

Fig. 18. Integration of GDT models with 3D point clouds and target inspection surfaces: (a) Office building, (b) Steel truss bridge.

Table 1
Time required to construct GDT models using Polyfit framework [49].

Building		Bridge	
Workflow	Time(sec)	Workflow	Time(sec)
RANSAC	1.4	RANSAC	0.16
Face hypothesis	1.63	Face hypothesis	0.14
Face confidences	17.77	Face confidences	2.4
Face optimization	16.4	Face optimization	2.01
	Total: 37.2		Total: 4.71

Table 2
Comparison of GDT model with other models.

	GDT Model	Scan to BIM	Manually BIM	Manually 3D model
Pre-requisites	Point clouds	Point clouds	Design drawings	Point clouds
Time	10–40(sec)	10+ hours [28] (including training)	A few hours	5–35(min)
Efficiency	High	Low	Low	Medium
Modeling method	Automatic	Automatic	Manual	Manual

Table 3
Calculation results of inspection waypoint parameters.

Objection	W_{sens} (mm)	H_{sens} (mm)	d (m)	f (mm)	W_{FOV} (m)	H_{FOV} (m)
Building	17.3	13.0	5	12.29	7.04	5.28
Bridge	17.3	13.0	9	12.29	12.67	9.52
Objection	C_x (m)	C_y (m)	D_{px} (m)	D_{py} (m)	Image size (px)	Resolution (mm/px)
Building	0.7	0.5	6.34	4.78	5280 × 3956	1.33
Bridge	1.2	0.9	11.47	8.62	5280 × 3956	2.40

Table 4
Statistics of number of inspection waypoints.

		Building	Bridge
Generated inspection waypoints		98	39
Adjusted inspection waypoints	unmoved	43	17
	moved	55	22
	added	1	0
Transitional waypoints		63	53
Repeat waypoints		11	0
Total number of waypoints		98 + 1–11 + 63 = 151	39 + 0–0 + 53 = 92
Total number of inspection images		98 + 1 = 99	39 + 0 = 39

inspection if the distance between the voxel and inaccessible voxels is less than d. The third category is the safe voxels, which are considered safe for UAV if the center point of the voxel is outside the GDT model and the distance to inaccessible voxels exceeds d. The VOI is the result of voxel classification, which expresses the position and geometric shape of obstacles.

Secondly, based on the VOI, risky waypoints located in inaccessible voxels or unsafe voxels can be identified. Subsequently, these risky waypoints are moved to the nearest safe voxel since each waypoint corresponds to a unique FOV centroid, even if multiple risky waypoints are adjusted to the same safe voxel, they still point towards their respective FOV centroids. Therefore, this adjustment does not create issues with image overlap. If multiple nearest safe voxels exist, the risky waypoints are prioritized to move along the outward normal direction. This adjustment strategy ensures that all waypoints are safe.

Finally, since the positions of some waypoints have been adjusted, altering the status of the full cover waypoints generated in Section 2, it is crucial to ensure that the waypoints, after being moved, can still fully cover the structural surfaces. Therefore, it is necessary to re-evaluate the image coverage to ensure that no inspection areas are missed. The process of calculating each camera's FOV coverage structural surface area is shown in Fig. 14. First, the GDT model surfaces are subdivided into dense discrete points using the grid mapping method proposed in Subsection 2.2.2 with parameters contain the horizontal and vertical spacing between dense points, D_{dx} and D_{dy}, should be 1 % D_{px} and D_{py}, respectively. This method not only ensures that dense discrete points accurately represent the coverage area of the FOV ensuring less than 1 % coverage error but also maintains computational efficiency. Subsequently, the camera's FOV for adjusted waypoints is re-calculated to check whether any discrete points are not covered by the camera's FOV based on the adjusted waypoints, ensuring that the surface of the GDT model has been fully covered by the UAV camera. It should be noted that UAV camera typically have a small focal length, providing a wide field of view and deep depth of field, which generally prevents blurring of the structural surface images. The proposed waypoint adjustment method can compensate the adverse effects of position changes on resolution while ensuring safety. Additionally, adjusting the focal length can further mitigate this impact. Whether discrete points are covered is determined by the angle between the normal vector of the discrete points and the view line, as well as whether there are any obstructions along the view line reaching the point. Additionally, the discrete points are consistent with the structure's surface. Therefore, it is appropriate to evaluate whether the adjusted waypoints fully cover the surface's discrete points using the accurate normal vectors provided by the GDT model. If there are uncovered discrete points, cluster these points to identify the uncovered areas. For each uncovered area, extract the center of the area as an additional FOV centroid. Correspondingly, an additional waypoint needs to be added; this waypoint should in the safe area along the outward normal direction of the model surface. This process continues until the UAV camera's FOV covers all discrete points.

By adjusting the preliminary generated waypoints based on the proposed VOI method, comprehensive coverage inspection of the structure can be achieved, while ensuring the safety of the UAV during inspection.

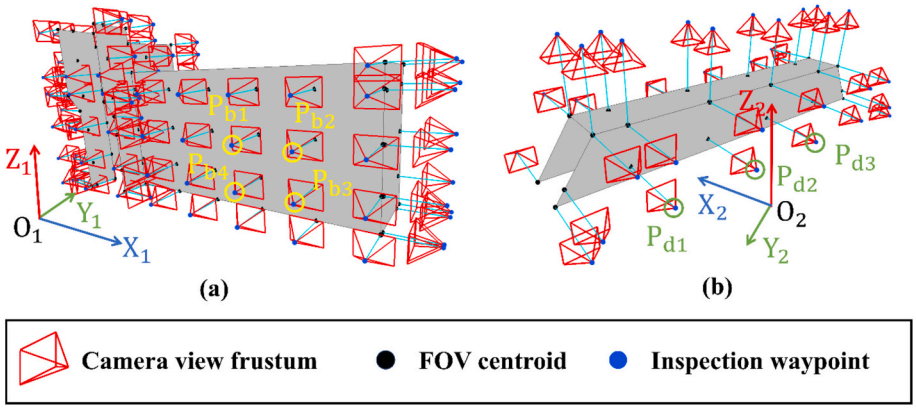


Fig. 19. Preliminarily generated inspection waypoints: (a) Waypoints of building, (b) Waypoints of bridge.

Table 5
Verification of consistency in waypoints generation and parameter settings.

Objection	Waypoints coordinates	Result distance between waypoints	Setting distance
Building	P_{b1}(15.888, 6.494, 53.915)	1–2: 6.34 m	6.34 m
	P_{b2}(22.227, 6.582, 53.915)	3–4: 6.34 m	6.34 m
	P_{b3}(22.227, 6.630, 49.136)	1–4: 4.78 m	4.78 m
	P_{b4}(15.887, 6.542, 49.136)	2–3: 4.78 m	4.78 m
Bridge	P_{d1}(−1.941, 22.783, 38.001)	1–2: 11.47 m	11.47 m
	P_{d2}(−5.467, 11.868, 38.001)	2–3: 11.47 m	11.47 m
	P_{d3}(−8.993, 0.954, 38.001)	1–3: 22.94 m	22.94 m

2.4. Flight path optimization method combining GA with A based on VOI*

The flight path is defined as a route that sequentially connects all generated inspection waypoints. Due to the unordered nature of the waypoints generated during the preceding processes, employing the initial sequence to establish the flight path could result in collision risks and significantly elevated energy consumption during the inspection process. Consequently, optimizing the sequence of waypoint connectivity to minimize the total travel distance and adding appropriate transitional waypoints to alter the unsafe or inaccessible flight path between two waypoints becomes imperative. As mentioned in the Section 1, previous studies have not sufficiently considered the issue that flight paths should maintain a safe distance from environmental obstacles, and reduce the costs associated with UAV turns due to obstacle avoidance. Therefore, a heuristic path optimization method combining GA with A* based on VOI is proposed, which constrained by the VOI, flight direction and turns costs.

As shown in Fig. 15, the process is as follows: initially, based on the GDT model, it is easy to determine whether a line connecting two waypoints passes through the target structure. Therefore, using a GA constrained by this condition can optimize the shortest flight path that passes through all waypoints without crossing the target structure. Many current path planning studies use GA [5,22,23], where GA treats flight paths formed by different connection orders of waypoints as individuals in nature. By crossing and mutating operations, GA changes the connection order of the parent individual to generate new offspring with different connection orders. The offspring with higher fitness are selected to repeatedly generate new individuals. In this problem, fitness refers to the distance of the current flight path, where a shorter flight path distance corresponds to higher fitness of the individual. The shortest flight path optimization problem constrained by flight direction costs can be represented by the objective function Eq. (6). When optimizing this problem using GA, the objective function is negatively correlated with fitness, where GA generates the individual with the highest fitness by adjusting the waypoint arrangement order to

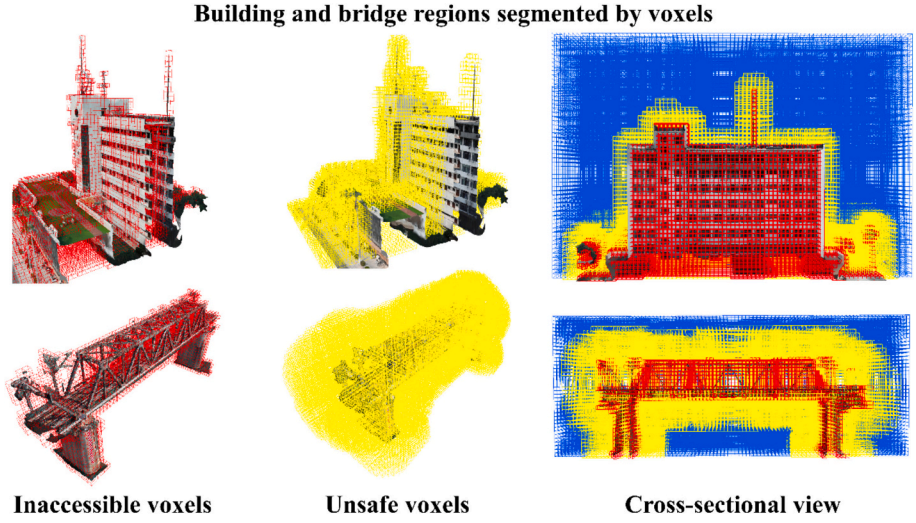


Fig. 20. Division of areas in UAV workspace by voxelization.

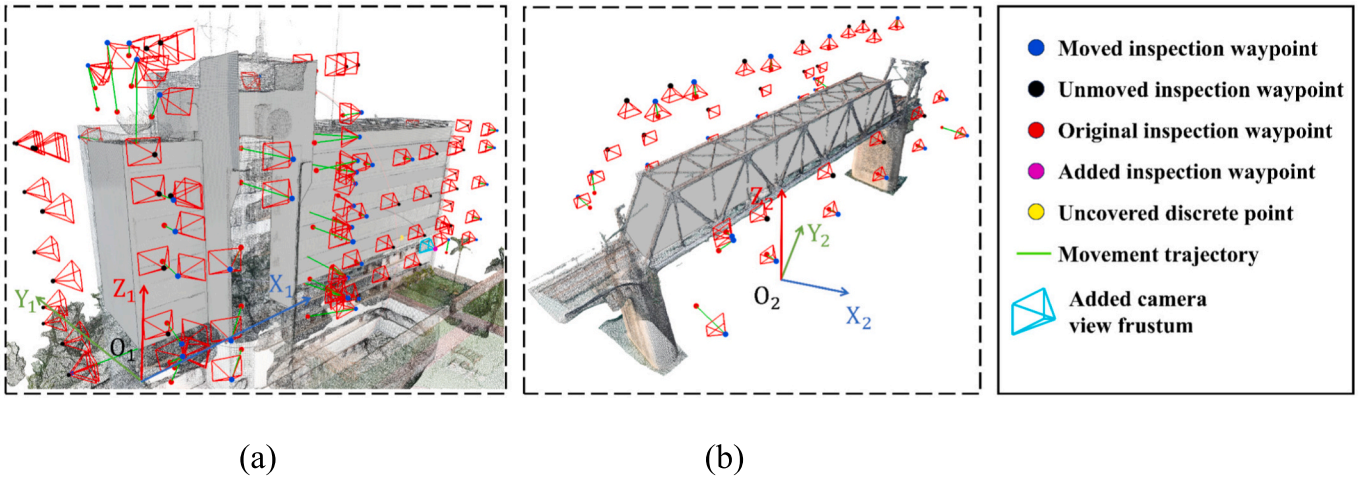


Fig. 21. Adjusted inspection waypoints: (a) Adjusted and added inspection waypoints for building, (b) Adjusted and added inspection waypoints for bridge.

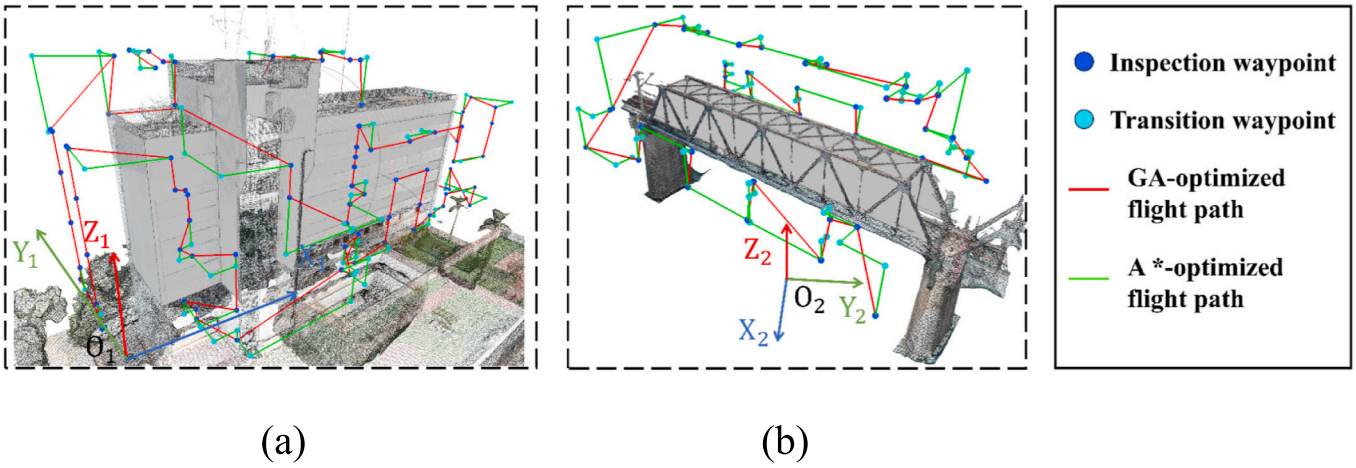


Fig. 22. Flight paths for executing UAV inspection mission: (a) Flight path for building, (b) Flight path for bridge.

minimize $f(P)$.

$$f(P)=\sum_{i=0}^{n-1}\lambda_i\left[\left(p_{i+1,1}-p_{i,1}\right)^2+\left(p_{i+1,2}-p_{i,2}\right)^2+\alpha\left(p_{i+1,3}-p_{i,3}\right)^2\right] \quad (6)$$

where, P is the flight path that passes through all waypoints, n is the number of waypoints, $p_{i,1}$, $p_{i,2}$ and $p_{i,3}$ are the first, second, and third components i-th waypoints in the flight path P, respectively, $p_{i+1,1}$, $p_{i+1,2}$ and $p_{i+1,3}$ are the first, second, and third components $i+1$-th waypoints in the flight path P, respectively, λ_i represents the distance weighting coefficient calculated through Eq. (7) and α represents the flight direction energy consumption ratio coefficient calculated through Eq. (8).

$$\lambda_i=\begin{cases}1\times10^5 \text{ if } p_{s,i}\in l(p) \text{ and } p_{s,i}\in R(P)\\ 1 \text{ if } p_{s,i}\notin l(p) \text{ or } p_{s,i}\notin R(P)\end{cases} \quad (7)$$

where, $p_{s,i}$ is the intersection point of the line connecting waypoint p_i and waypoint p_{i+1} with the plane of the GDT model surface, $l(p)$ represents the set of points within line segment between p_i and p_{i+1}, and $R(P)$ represents the set of points within the polygonal regions of the GDT model surface.

In Eq. (7), $p_{s,i}\notin l(p)$ means that although the line passing through the two waypoints intersects with the plane of the GDT model surface, the intersection point is not located between the two waypoints, and therefore, the surface does not create an obstruction. The other condition, $p_{s,i}\notin R(P)$, indicates that an intersection point exists, but it lies outside the boundary of the GDT model surface, so it does not create an obstruction either.

$$\alpha=\begin{cases}W_2/W_1 \text{ if } p_{i+1,3}-p_{i,3}>0\\ W_3/W_1 \text{ if } p_{i+1,3}-p_{i,3}<0\end{cases} \quad (8)$$

where, W_1 represents the motor power of the UAV during level flight at the set flight speed in calm wind environment, W_2 and W_3 represent the motor power during upward and downward flight under the same conditions, respectively.

Subsequently, as described by Tan et al. [23], the flight paths optimized by the GA connect waypoints with straight lines, thus unable to navigate around obstacles. After obtaining the shortest flight path that connects all waypoints and does not cross the GDT model using GA, the proposed VOI-based flight path safety assessment method is used to check if each segment of the flight path is safe. For unsafe or inaccessible segments, the A* algorithm is employed to find transitional waypoints to alter the original straight-line connections, ensuring the flight path avoids and maintains a safe distance from environmental obstacles. The VOI-based flight path safety assessment method involves discretizing the flight path between two waypoints by a distance d_s, where the choice of d_s affects the computational speed and accuracy of the safety assessment. This paper considers a reasonable value for d_s to be $0.2S$, where S is the voxel size in VOI (as mentioned in Subsection 2.3). If all discrete points lie within safe voxels, the flight path segment between two waypoints is deemed safe; otherwise, the A* algorithm is required to find a safe path to alternate origin segment for the two waypoints

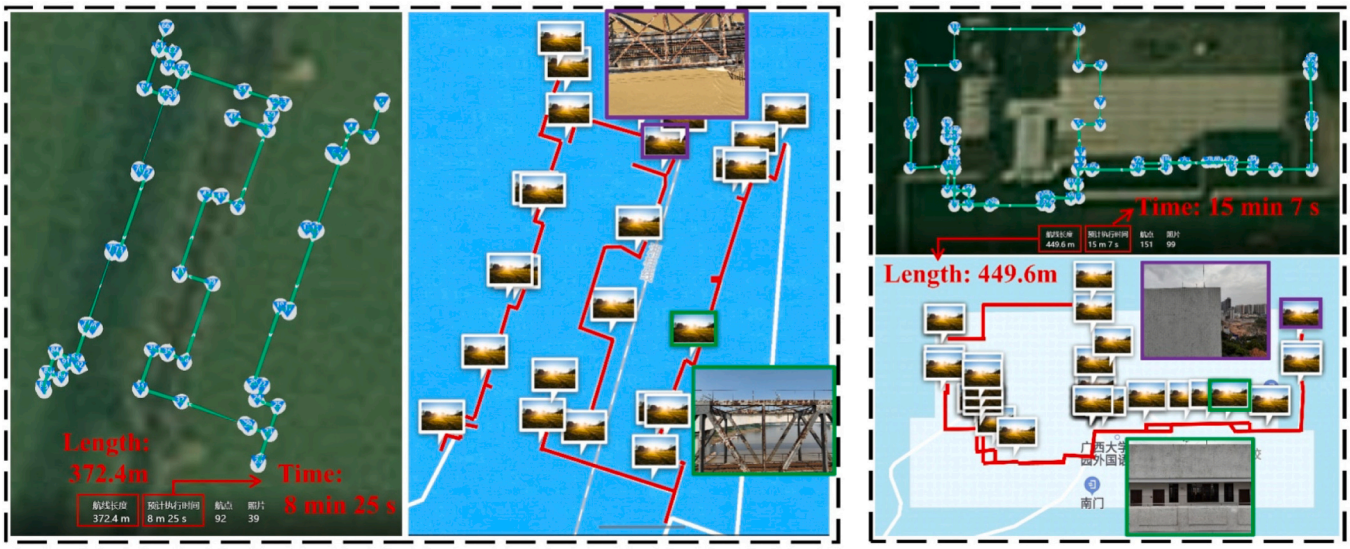


Fig. 23. Flight path generated by flight mission file and actual flight path of UAV: (a) Flight paths for bridge inspection, (b) Flight paths for building inspection.

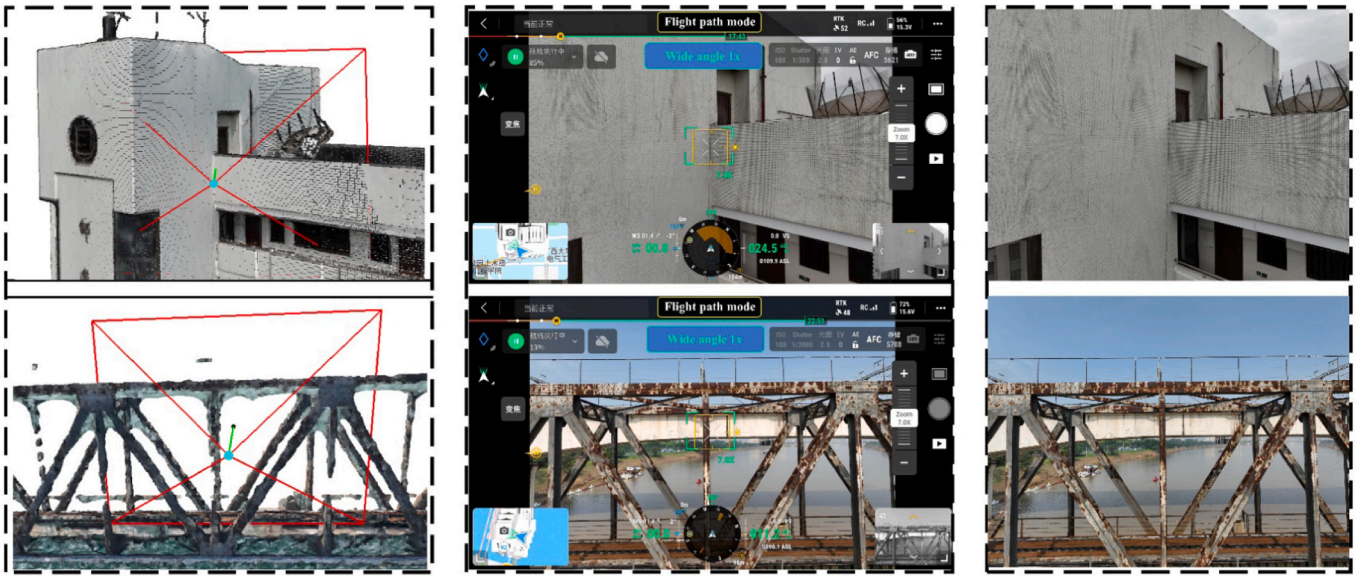


Fig. 24. Inspection images acquirement: (a) Expected inspection images from planned waypoints, (b) Screenshots of remote controller during mission execution, (c) Inspection images captured during mission.

connecting. Considering that the UAV needs to decelerate when approaching waypoints to prepare for the next action, either moving to the next waypoint or capturing images, and needs to accelerate when heading to the next waypoint to reach the set flight speed, using nodes found by the A* algorithm as transitional waypoints could lead to frequent acceleration and deceleration, thus consuming more energy. Nodes located on the same straight line may not be entered as transitional waypoints, thereby reducing the number of direction changes in the path found by the A* algorithm and minimizing the number of transitional waypoints, making the UAV operation smoother and more energy-efficient. Therefore, based on Tan et al. [23], the objective function for the A* algorithm, considering turning penalty and based on VOI, is described in Eqs. (9)–(13).

$$f(n_i) = \gamma_i(n_i) + g(n_i) + h(n_i) \tag{9}$$

where, $f(n_i)$ represents the total cost of the current node n_i, $\gamma_i(n_i)$ represents the safety cost of the current node n_i, determined by Eq. (10); $g(n_i)$ represents the cost from the starting waypoint to the current node n_i, calculated by Eq. (11) and Eq. (12); $h(n_i)$ represents the distance cost from the current node n_i to the end waypoint, calculated by Eq. (13).

Table 7
Camera orientation adjustment parameters during bridge inspection.

NO.	Inspection waypoints in GCS			FOV centroids in GCS			Waypoints in WGS-84	
	X	Y	Z	X	Y	Z	Lat	Long
1	27.28	22.61	84.87	14.62	26.70	85.34	22.80325	108.29015
2	27.29	22.70	91.67	–	–	–	22.80325	108.29016
…	…	…	…	…	…	…	…	…
92	−14.35	−15.22	92.00	−5.78	−17.99	92.21	22.80291	108.28975

NO.	Target vector $\overrightarrow{T}$	Initial vector $\overrightarrow{V}$	$\overrightarrow{T}_1$	Rotation angle	Pitch angle	Speed
1	(−12.66, 4.09, 0.47)	(0, 1, 0)	−12.66	−72.11°	2.04°	3 m/s
2	–	–	–	–	–	3 m/s
…	…	…		…	…	…
92	(8.56, −2.77, 0.21)	(0, 1, 0)	8.56	107.90°	1.36°	3 m/s

$$\gamma_i(n_i) = \begin{cases} 1 \times 10^5 \text{ if } n_i \notin S(P) \\ 0 \text{ if } n_i \in S(P) \end{cases} \tag{10}$$

where, n_i represents the current node; $S(P)$ represents the set of center points of safe voxels in the VOI.

$$g(n_i) = g(n_{i-1}) + (1 + \delta_i)S \tag{11}$$

where, $g(n_{i-1})$ represents the cost from the starting waypoint to the previous node n_{i-1}; S is the size of the voxels in VOI, and since the nodes are the center points of voxels, the distance between nodes is equal to the size of the voxels; δ_i represents the path turning penalty coefficient, calculated by Eq. (12).

$$\delta_i = \begin{cases} 1 \text{ if } (n_{i-2} - n_{i-1}) \bullet (n_{i-1} - n_i) = 0 \\ 0 \text{ if } (n_{i-2} - n_{i-1}) \bullet (n_{i-1} - n_i) = 1 \end{cases} \tag{12}$$

where, n_{i-1} represents the coordinates of the node preceding n_i; n_{i-2} represents the coordinates of the node preceding n_{i-1}.

$$h(n_i) = |x_e - x_i| + |y_e - y_i| + |z_e - z_i| \tag{13}$$

where, (x_e, y_e, z_e) represents the coordinates of the end waypoint; (x_i, y_i, z_i) represents the coordinates of the current node.

2.5. File compilation method for UAV inspection mission

The waypoints and flight path generated through the above methods enable the UAV to reach each inspection waypoint along the flight path. However, it is necessary to control the UAV's camera gimbal to face the target structure to capture the correct inspection images. As shown in Fig. 16(a), the control process is divided into rotate and pitch adjustments. The angle between the target vector $\overrightarrow{T}$ and the initial vector $\overrightarrow{V}$ can be decomposed into a rotation angle and a pitch angle, which are achieved by the differential rotational speed of the UAV's diagonal propellers and the adjustment of the camera gimbal, respectively. If the UAV is not directly above the FOV centroid, the vector between them can be used as $\overrightarrow{T}$ to calculate the rotation angle and pitch angle. Otherwise, the vector between the UAV and the FOV centroid is orthogonal to the XOY plane, making it impossible to decompose the angle. In this case, the vector of the longest vertical edge on the top surface is used as $\overrightarrow{T}$ to calculate rotation angle, ensuring that the width of images captured by the UAV are parallel to the longest edge on the top surface, facilitating better inspection results. The default camera direction vector of the DJI UAV points north, and the default pitch angle of the gimbal is 0 degrees, therefore the initial vector in GCS is (0,1,0). The rotation angle and pitch angle can be calculated by Eqs. (14)–(16).

$$\overrightarrow{T} = \begin{cases} P_{FOV} - P_I \text{ if } arccos\left|\dfrac{(P_{FOV} - P_I) \bullet (0,0,1)}{\|(P_{FOV} - P_I)\|_2}\right| \neq 0 \\ \overrightarrow{n} \times \overrightarrow{l}_{max} \text{ if } arccos\left|\dfrac{(P_{FOV} - P_I) \bullet (0,0,1)}{\|(P_{FOV} - P_I)\|_2}\right| = 0 \end{cases} \tag{14}$$

where, $\overrightarrow{T}$ represents the target vector for acquire correct inspection images; P_{FOV} represents the coordinates of the FOV centroids in GCS; P_I represents the coordinates of the Inspection waypoints in GCS; $\overrightarrow{n}$ is the normal vector of the polygonal plane of the GDT model; $\overrightarrow{l}_{max}$ is the vector of the longest edge on the surface of the GDT model.

$$\angle R = \frac{\overrightarrow{T}_1}{\left|\overrightarrow{T}_1\right|} \times arccos\frac{\left(\overrightarrow{T}_1, \overrightarrow{T}_2\right) \bullet (0,1)}{\left\|\left(\overrightarrow{T}_1, \overrightarrow{T}_2\right)\right\|_2} \tag{15}$$

where, $\angle R$ represents the rotation angle needs to be adjusted, if $\overrightarrow{T}_1 > 0$, $\frac{\overrightarrow{T}_1}{\left|\overrightarrow{T}_1\right|} = 1$, the angle value is positive, it indicates a clockwise rotation from the initial direction, if $\overrightarrow{T}_1 < 0$, $\frac{\overrightarrow{T}_1}{\left|\overrightarrow{T}_1\right|} = -1$, it is negative, it indicates a counterclockwise rotation from the initial direction, if $\overrightarrow{T}_1 = 0$, $\angle R = 0$; $\overrightarrow{T}_1$ and $\overrightarrow{T}_2$ are the first and second components of vector $\overrightarrow{T}$ (respectively)

$$\angle P = \begin{cases} arctan\dfrac{\overrightarrow{T}_3}{\left\|\left(\overrightarrow{T}_1, \overrightarrow{T}_2\right)\right\|_2} \text{ if } arccos\left|\dfrac{(P_{FOV} - P_I) \bullet (0,0,1)}{\|(P_{FOV} - P_I)\|_2}\right| \neq 0 \\ \dfrac{(P_{FOV} - P_I) \bullet (0,0,1)}{|(P_{FOV} - P_I) \bullet (0,0,1)|} \times \dfrac{\pi}{2} \text{ if } arccos\left|\dfrac{(P_{FOV} - P_I) \bullet (0,0,1)}{\|(P_{FOV} - P_I)\|_2}\right| = 0 \end{cases} \tag{16}$$

where, $\angle P$ represents the pitch angle needs to be adjusted, $\overrightarrow{T}_3$ is the third component of vector $\overrightarrow{T}$.

Based on the above analysis, the following information can be compiled into a UAV flight mission file [see Fig. 16(b)]: the geographic coordinates of the inspection waypoints. As mentioned in Subsection 2.4, the A* algorithm generates transitional waypoints to navigate the flight path around obstacles. Transitional waypoints will not capture images and will be directly compiled into the flight mission file with their coordinates, whereas inspection waypoints that require image capture will need more detailed preparation. This includes calculating the rotation and pitch angle information based on the vector from the inspection waypoints to the corresponding FOV centroids relative to the initial vector, as well as the camera action information for hovering and adjusting the focus to capture images after the UAV is oriented towards

the target. Once the file is compiled, it can be output in a standard format and imported into flight control software to enable the UAV to automatically execute the inspection mission.

3. Field experimental validation

To verify the effectiveness of the proposed automatic planning method for UAV inspection missions, it was applied to some representative real-life structures, i.e., an 8-storeyed office building [Fig. 17(a)] and a 56 m-span steel truss bridge [Fig. 17(b)]. The UAV used to collect multi-view images of the structures and execute the inspection missions was a DJI Mavic3 Enterprise model, which has a maximum battery life of 46 min. It is equipped with an RTK module that enhances the positioning capability of the UAV, and features a wide-angle camera sensor size of 4/3 with 20 million effective pixels. The computer used for modeling and running the algorithms was equipped with an Intel Core i7–10,700 CPU, an Nvidia RTX 3080 12G GPU, and 32GB of RAM.

3.1. 3D point clouds reconstruction

The selected structures are reconstructed into 3D point clouds using the multi-view 3D reconstruction method, with iTwin Capture utilized for this 3D reconstruction. To simplify the GDT model construction and omit some unnecessary details, a subset of multi-view images is used to reconstruct a rough 3D point cloud (as shown in Fig. 18). In contrast, the complete set of multi-view images is employed to generate a refined 3D point cloud (as shown in Fig. 20) for accurate VOI generation. To ensure the correct display of the 3D point clouds, the GCS is chosen for exporting the reconstructed 3D point clouds. Subsequently, the generated inspection waypoints need to be imported into the flight control system using the WGS-84 coordinate system. The conversion between these two coordinate systems is facilitated by a MATLAB program developed specially for this purpose. The reconstruction results for the building and bridge 3D point clouds are shown in Fig. 6.

3.2. Inspection flight paths planning

The output 3D point clouds from 3D reconstruction are input into the Polyfit framework [49] for GDT model construction. As shown in Fig. 18, the models closely align with the 3D point clouds, accurately representing the structural geometric attributes. Some representative structural surfaces will be selected as targets for automatic inspection. According to Eq. (1), the RMSEs between the 3D point clouds and the GDT models of the building and bridge are calculated to be 0.29 m and 0.18 m, respectively. The volumes of the building and bridge are approximately 23,972 m^3 and 2540 m^3, respectively. As mentioned in Subsection 2.1, the calculated RMSE values normalized by L are 0.010 L and 0.013 L, respectively. Considering both the RMSE results and the alignment shown in Fig. 18, the GDT models accurately represent the structures.

The time required to input 3D point clouds into the Polyfit framework [49] for constructing GDT models is shown in Table 1. A comparison of the GDT models with other models is presented in Table 2. For both the building and the bridge, automatic modeling takes less than one minute, significantly enhances modeling efficiency compared to other methods. Moreover, except for inputting the RANSAC parameters, the modeling process is algorithmically handled without the need for manual intervention.

Subsequently, the waypoint parameters are calculated, with results presented in Table 3. The working distance d is determined based on the analysis in Subsection 2.2. As demonstrative applications, the inspection images will be used to inspect cracks on the exterior walls for the building, and considering the complex surrounding environment, d is set to 5 m. For the bridge, the inspection images will be used to check for extensive corrosion, and given its more open surrounding environment, d is set to 9 m. To compensate the adverse impact of UAV positioning errors on inspection image coverage, the image overlap distance is set to 10 % of the waypoint spacing and rounded down. After calculating the waypoint parameters, inspection waypoints are generated using the method introduced in Subsection 2.2, with the results shown in Fig. 19 and their number listed in Table 4. The grid mapping algorithm and parallel shift algorithm were written in Python 3.10 and executed using PyCharm. The effectiveness of the generated inspection waypoints was validated by comparing the coordinates of the partial generated waypoints (as shown in Fig. 19) with the waypoint parameters, the validation results are presented in Table 5.

After generating the inspection waypoints using the method described in Subsection 2.3, the bounding box of the 3D point cloud is expanded to define the working space of the UAV, with expansion distances d_w of 7 m for the building and 11 m for the bridge. The voxel size S set to 1 m. Subsequently, based on the 3D point clouds and the GDT model, the VOI is obtained, and the area of the working space is divided into inaccessible, unsafe, and safe areas (as illustrated in the cross-sectional view in Fig. 20). Based on the VOI, the inspection waypoints are adjusted, as shown in Fig. 21 and their number listed in Table 4, the proportion of unsafe waypoints are 56.1 % and 56.4 %, respectively. Subsequently, the coverage of the GDT model surface by the adjusted inspection waypoints is re-evaluated using the method in Subsection 2.3. Corresponding to waypoint generation, UAV positioning errors also need to be considered during the re-evaluation process, with the FOV coverage area reduced by a ratio defined as the proportion of the C_x and C_y to D_{px} and D_{py}, respectively. Upon re-evaluation, it was found that for the building, the adjusted inspection waypoints did not fully cover the GDT model surface. The uncovered areas are shown as yellow regions in Fig. 21(a). To achieve full coverage, additional inspection waypoints were added for these uncovered areas. Notably, the Repeat waypoints in Table 4 refer to situations where multiple inspection waypoints are moved to the same safe voxel. In such cases, these waypoints count as a single inspection waypoint. When the UAV reaches this waypoint, it will adjust to multiple angles to capture several inspection images.

After obtaining all waypoints, optimize the flight path using the Flight path optimization method combining GA with A* based on VOI proposed in Subsection 2.4. First, GA constrained by the GDT model and flight direction energy consumption is used to optimize the shortest flight path [As shown in Fig. 22(a)]. Based on the motor power consumption during flights in different directions in a windless environment, the energy consumption ratios α for upward and downward flights are found to be 1.05 and 0.95, respectively. At this point, the path cannot avoid environmental obstacles. To address this, the A* algorithm based on VOI is used to perform a secondary optimization of the flight path. In calculating the cost, turning penalty δ_i is set to 2. Add as few transitional waypoints as possible to unsafe or inaccessible segments to avoid obstacles and ensure the safety of the flight path while minimizing the additional energy consumption caused by obstacle avoidance [As shown in Fig. 22(b)]. The UAV will follow the A*-optimized flight path to reach each inspection waypoint and capture inspection images.

3.3. Deployment of automatic inspection flight mission using compiled file

Based on the strategy outlined in Subsection 2.5 for adjusting the UAV's camera orientation, the coordinates of the inspection waypoints are matched with the FOV centroids they represent. The UAV's rotation angle and the gimbal's pitch angle required for camera adjustment are calculated by Eqs. (14)–(16), results are shown in Table 6 and Table 7 listed in the Appendix.

The information in Table 6 and Table 7 are then automatically compiled into KML files using the ET library in Python. Fig. 25 in the Appendix indicates that the calculated angles are consistent with the camera adjustment angles during actual flights. After organizing it into a flight mission file per DJI's standard format, it is imported into the DJI Pilot 2 software to deploy the inspection mission and control the UAV for automated inspection. The executed flight path and the automatically

captured inspection images are shown in Fig. 23 and Fig. 24. In Fig. 23, the red lines represent the path flown by the UAV during the automated inspection. It can be observed that the executed flight path aligns with the planned flight path. Fig. 24 shows inspection images captured from some of the inspection waypoints. By comparing the recorded coordinates of the UAV during image capture with the coordinates of the designated waypoints, the horizontal and vertical positioning errors for each waypoint were obtained. Fig. 26 in the Appendix presents the error results. The FOV errors corresponding to the maximum horizontal and vertical errors did not exceed the predetermined image overlap distances C_x and C_y. Additionally, using the side and top surfaces of the bridge case as examples, the image coverage of the structural surface was examined (as shown in Fig. 27 in Appendix). The structure in the images was stitched together based on feature points, covering the entire surface, indicating that the images can achieve full coverage of the structural surface. It can be observed that the objects contained in the captured images and their positions in the images align with the FOV of the planned inspection waypoints. This indicates that the proposed automatic structural surface inspection method is accurate and that the captured images meet the expected criteria. Additionally, the fact that the inspection images are parallel to the surface of the structure is consistent with the FOV of the unmoved inspection waypoints being parallel to the surface of the GDT model. This demonstrates that the proposed method for calculating the UAV's rotation and pitch angles is correct, allowing for the accurate adjustment of the UAV's orientation to capture the planned inspection images. The accuracy of the camera's FOV can be affected by waypoints error caused by the limitations of current positioning method, compass interference during mission execution, and 3D reconstruction inaccuracies. However, as mentioned in Subsection 2.2.1 and demonstrated by the full coverage evaluation results, these negative effects can be compensated by appropriately adjusting the waypoint generation and parameters. To validate the effectiveness of the inspection images, they were reviewed by experts. As shown in Fig. 28 in Appendix, the damage features are quite evident, allowing for easy identification and marking of the damage within the images.

4. Conclusions

This paper presented a UAV automatic inspection method for structural surface based on GDT model and VOI. After conceptual analysis and validation on two actual structures (a building and a bridge), the proposed method has demonstrated the potential for higher efficiency, lower collision risks, and greater automation level in extensive and diverse structural surface inspection. The main conclusions are as follows:

(1) To address the issue of low automation and efficiency in the generation of UAV inspection waypoints, based on the automatically constructed GDT model, the FOV centroids' parallel shift method for generating inspection waypoints is proposed. This method includes waypoint parameters calculation method, grid mapping method of dividing FOV for obtaining the centroids, and generating inspection waypoints based on FOV centroids' parallel shift. Compared to existing methods, the proposed method enhances automation, efficiency, and accuracy in generating inspection waypoints.

(2) To address the inspection waypoints' safety risk issue of insufficient detail, low accuracy, and weak robustness in considering environmental obstacles during UAV inspection process, a method for adjusting inspection waypoints based on environmental VOI of 3D point clouds is proposed. This method effectively divides safe working areas and adjusts waypoints not located in safe areas. Compared to approaches that do not consider obstacle avoidance, the results from two examples in this paper demonstrate that the proposed method reduces the number of risky waypoints from 55 and 22 to 0, respectively. Additionally, the risk of collision has been reduced drastically, significantly enhancing the safety of inspection waypoints.

(3) To address the issue of high energy consumption during flight and flight path safety risks, a flight path optimization method combining GA with A* based on VOI is proposed. This method first uses GA constrained by the GDT model and flight direction energy consumption to optimize the flight path, ensuring the shortest flight path and thus low energy consumption for the UAV's automatic inspection. Secondly, it employs A* based on VOI and turning penalty for secondary optimization, altering the parts of the path with safety risks to avoid real environmental obstacles. The field experiments of optimized flight paths added 63 and 53 transitional waypoints, respectively. This ensures the safety of the flight path. After waypoint adjustment, the inspection waypoints are guaranteed to be safe. After path optimization, the flight path is guaranteed to be safe. Ultimately, this ensures the UAV's safety during the inspection mission, reducing collision risks.

However, the proposed method still has limitations and can be improved in future research: (1) The GDT model only provides information about the external surface of the structure, which is sufficient for external inspections. However, for internal surface inspections, it does not assist in the specific inspection of areas like columns and beams on the structural surface. In the future studies, integrating additional information such as 3D reconstructed images and design drawings into the GDT model could enable more comprehensive and detailed inspections of the entire structure. (2) The automatic framework for establishing the GDT model works well for geometrically simple building structures and bridge structures. For more complex structures with curved surfaces, overhangs, other small structural details. New modeling methods will be developed for these structures in the subsequently studies. (3) The proposed waypoint adjustment method minimizes the changes to the positions of unsafe waypoints, limiting the impact of position changes on resolution. In the future, camera zooming could be utilized to further reduce this impact. (4) There are numerous factors affecting UAV energy consumption, and further research need be conducted on methods to reduce energy consumption through path optimization or improve the UAV flying mode, which will be one of the objectives of the subsequent studies. Relevant codes and data in the paper will be made available on reasonable request.

CRediT authorship contribution statement

Yonghui An: Writing – original draft, Formal analysis, Conceptualization, Methodology, Funding acquisition, Supervision, Resources. **Jianren Ning:** Writing – original draft, Visualization, Investigation, Formal analysis, Methodology, Data curation. **Chuanchuan Hou:** Writing – review & editing, Methodology, Data curation, Funding acquisition, Conceptualization, Supervision. **Jinping Ou:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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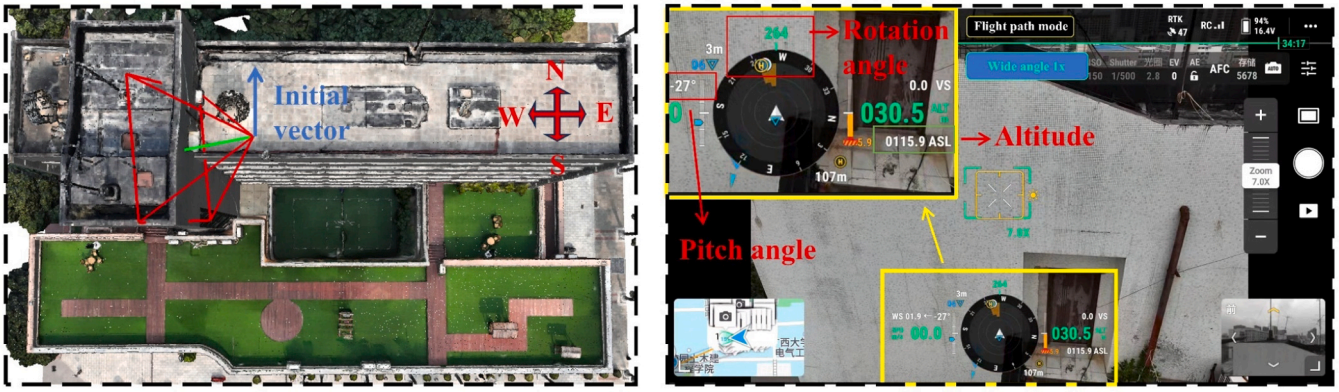
Appendix A. Field experimental calculation results

Some of the calculation and validation results are presented in the Appendix for readers' reference.

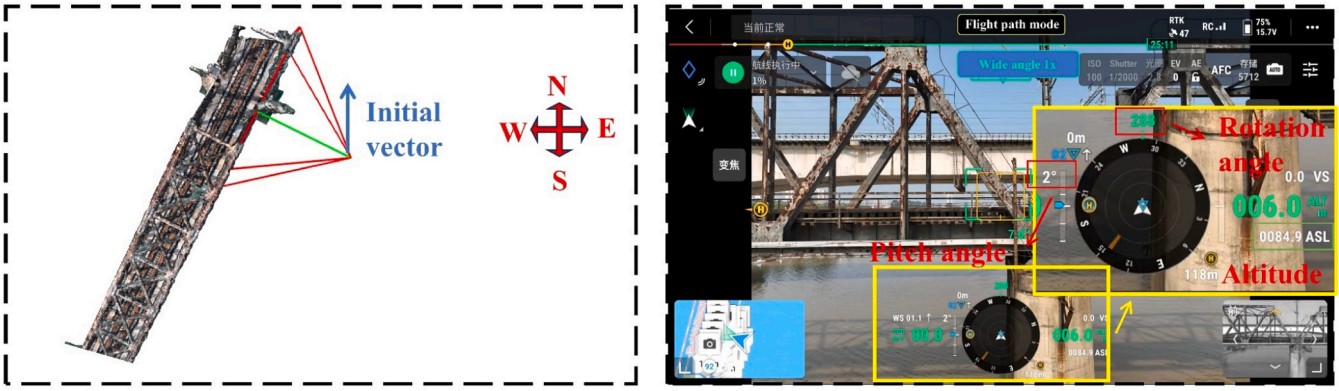
Table 6
Camera orientation adjustment parameters during building inspection.

NO.	Inspection waypoints in GCS			FOV centroids in GCS			Waypoints in WGS-84	
	X	Y	Z	X	Y	Z	Lat	Long
1	1.31	5.67	111.98	−3.43	7.67	111.64	22.83977	108.28407
2	1.31	5.67	115.98	–	–	–	22.83977	108.28407
…	…	…	…	…	…	…	…	…
5	1.31	11.67	115.98	−3.58 −3.58	11.11 11.10	113.46 115.33	22.83983	108.28407
…	…	…	…	…	…	…	…	…
151	−15.69	9.67	114.98	−10.59 −10.61	11.05 11.02	109.76 113.48	22.83981	108.28391

NO.	Target vector $\vec{T}$	Initial vector $\vec{V}$	$\vec{T}_1$	Rotation angle	Pitch angle	Speed
1	(−4.74, 2, −0.34)	(0, 1, 0)	−4.74	−67.12°	−3.78°	3 m/s
2	–	–	–	–	–	3 m/s
…	…	…	…	…	…	…
5	(−4.90, −0.56, −2.52) (−4.89, −0.58, −0.65)	(0, 1, 0) (0, 1, 0)	−4.90 −4.89	−96.50° −96.72°	−27.10° −7.54°	3 m/s
…	…	…	…	…	…	…
151	(5.09, 1.38, −5.23) (5.08, 1.34, −1.50)	(0, 1, 0) (0, 1, 0)	5.09 5.08	74.82° 75.16°	−44.71° −15.95°	3 m/s



(a)



(b)

Fig. 25. UAV camera angle adjustment: (a) Planned camera angle and actual UAV rotation angle for first inspection image captured at waypoint No. 5 in building task, corresponding to waypoint No. 5 in Table 6, (b) Planned camera angle and actual UAV rotation angle for inspection image captured at waypoint No. 1 in bridge task, corresponding to waypoint No. 1 in Table 7.

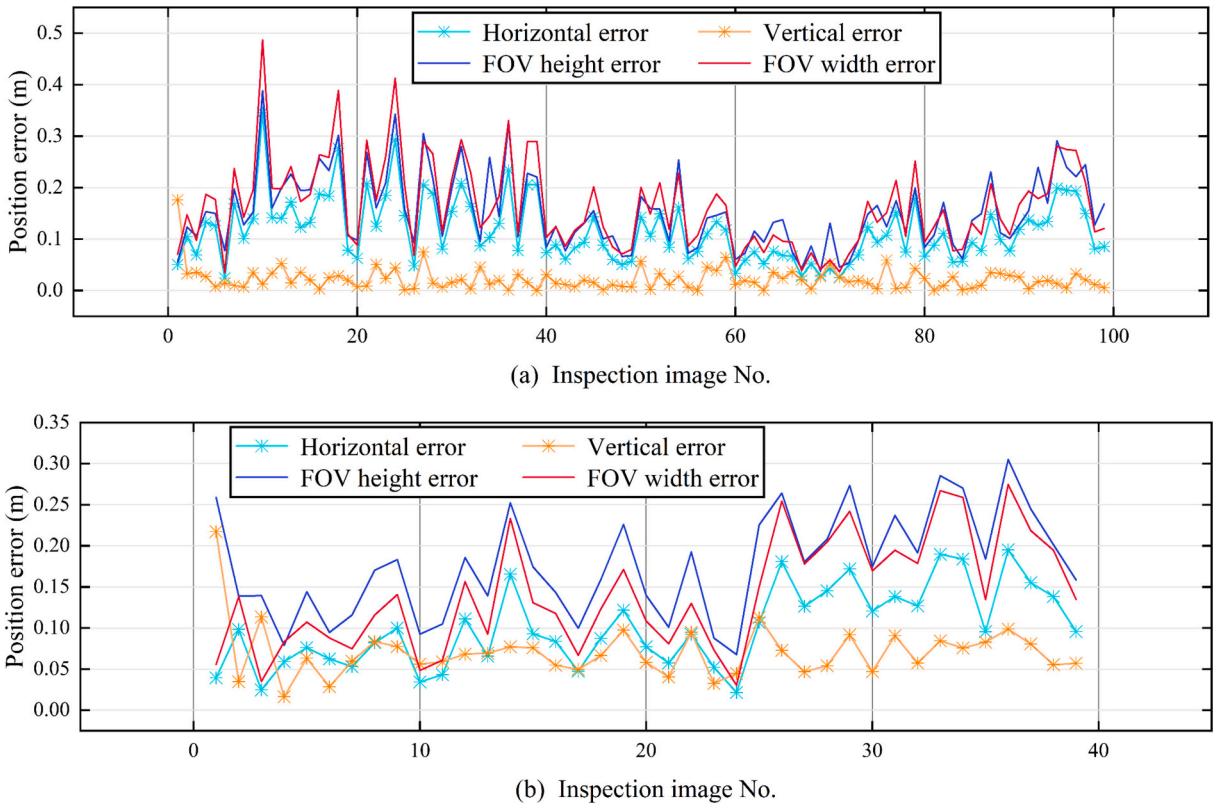


Fig. 26. Coordinate errors between inspection images and set waypoints and their impact on FOV coverage: (a) Coordinate and FOV errors for building example, (b) Coordinate and FOV errors for bridge example.

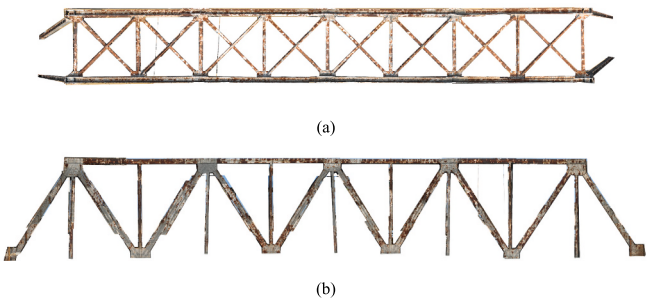


Fig. 27. Captured images for complete coverage inspection of bridge surface: (a) Stitched images of top surface, (b) Stitched images of side surface.

Fig. 28. Detected damage from images: (a) Corrosion on exterior surface of bridge structure, (b) Cracks on exterior surface of building structure.

Data availability

Data will be made available on request.

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